

For which patients shall the drug work, and how?

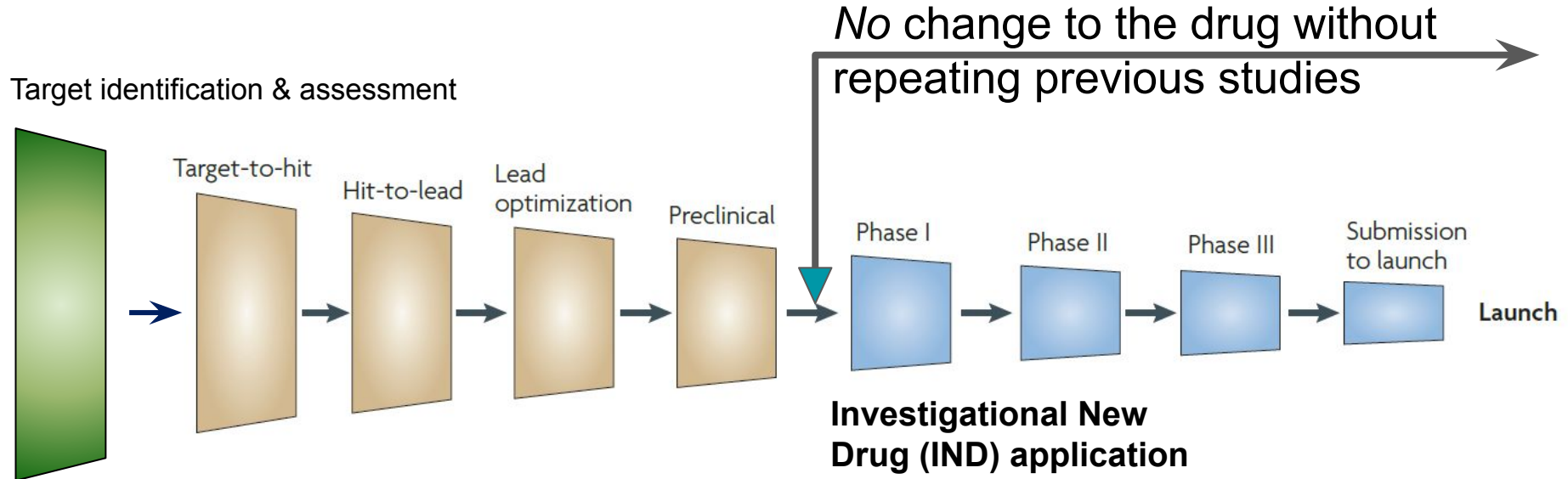
Mathematical and Computational Biology in Drug Discovery (MCBDD)
Module V

Dr. Jitao David Zhang
May 2025

Outline of Module V

- Lecture 11
 - Biomarker for dose prediction
 - Biomarker for patient-stratification and biology understanding: Merck/Genentech
 - Challenges and caveats
- Lecture 12
 - Integrating statistical and mechanistic modelling: Griffiths *et al.*
 - Mechanistic modelling of biological systems: from Boolean network to Agent-based modelling
 - Causal inference

From drug discovery to drug development



Phases of clinical trials

Investigational New Drug
(IND) application

New Drug Application (NDA)

Phase 0*

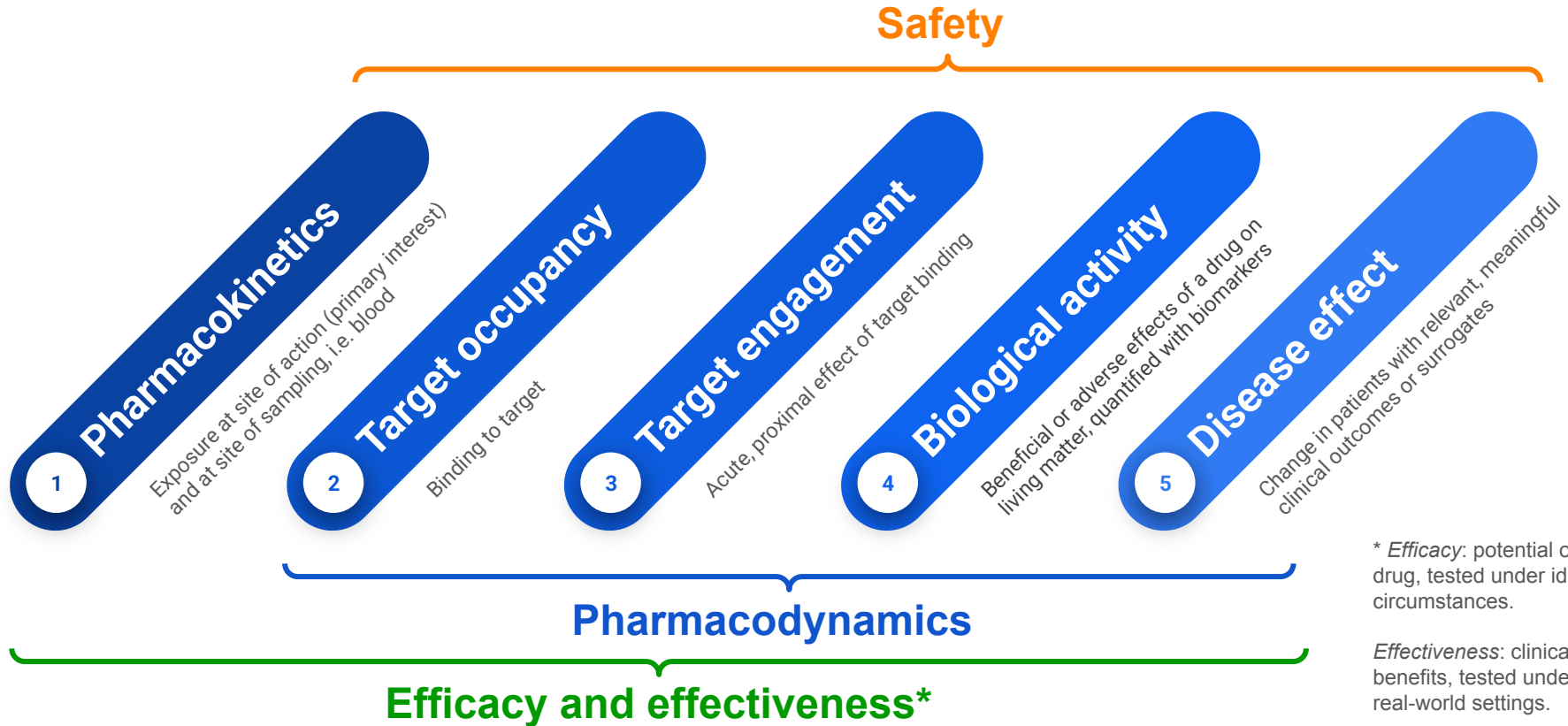
Phase I ~70%

Phase II ~50%

Phase III ~60%

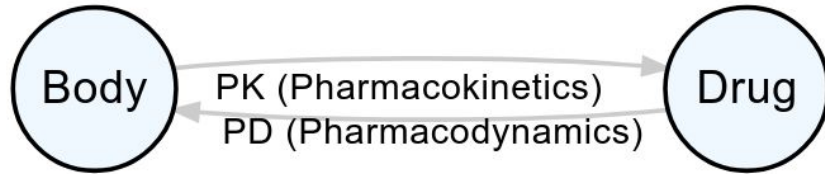
- **Aim:** Getting PK/PD data to verify the drug behaves as expected.
- **Dose:** *Microdosing, e.g. 1% of predicted dose.*
- **Subjects:** <15 healthy subjects
- **Time:** A few weeks
- Finding safe dose ranges and optimal dosing regimens with further PK/PD data.
- Sub-therapeutic single and multiple ascending doses
- 20-100 healthy subjects (patients)
- A few months
- Assessing efficacy and safety profiles of the drug, and determining the dosing regimen.
- Therapeutic dose
- Usually 100-300 patients with a specific disease
- A year or longer
- Comparing efficacy, effectiveness, and safety profiles with the standard-of-care treatment option.
- Therapeutic dose
- Usually 300-3000 patients
- Usually several years

The chain of translation

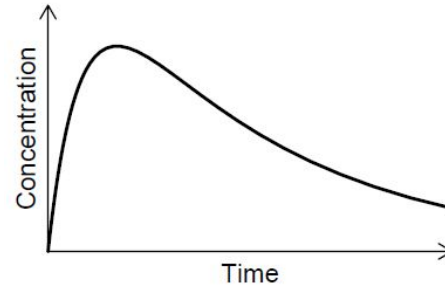


A refresher of PK/PD Modelling

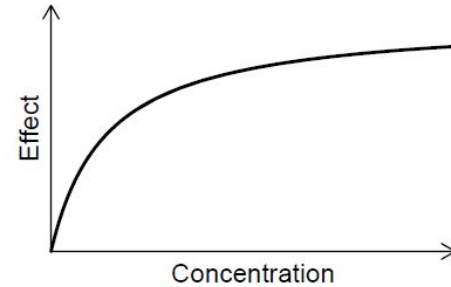
(A)



(C)

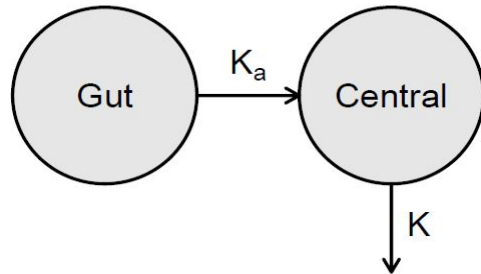


(a) PK model



(b) PD model

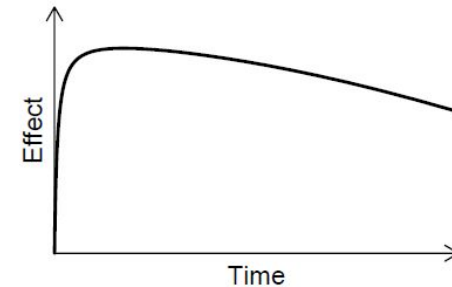
(B)



$$\frac{dA_{gut}}{dt} = -K_a \cdot A_{gut}$$

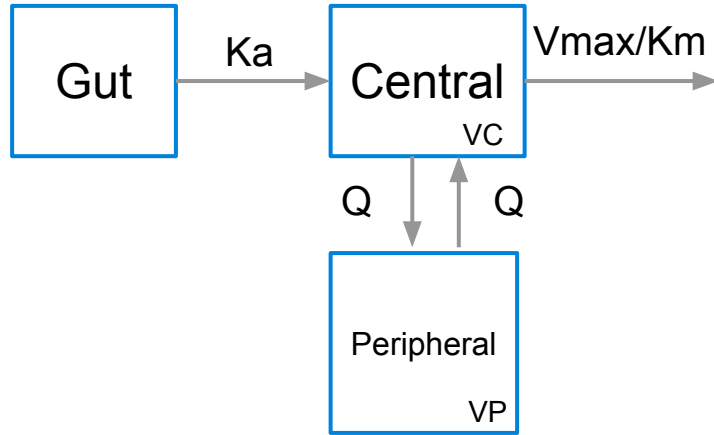
$$\frac{dA}{dt} = \underbrace{F \cdot K_a \cdot A_{gut}}_{\text{from gut}} - \underbrace{K \cdot A}_{\text{elimination}}$$

$$A_{oral}(t) = \frac{K_a F A_0}{K_a - K} (\exp(-K \cdot t) - \exp(-K_a \cdot t))$$



(c) Combined PK/PD model

An example of a two-compartment PK model



Ordinary Differential Equation (ODE) based model of *in vivo* PK, assuming two compartments (central and peripheral), and the Michaelis-Menten model of drug elimination.

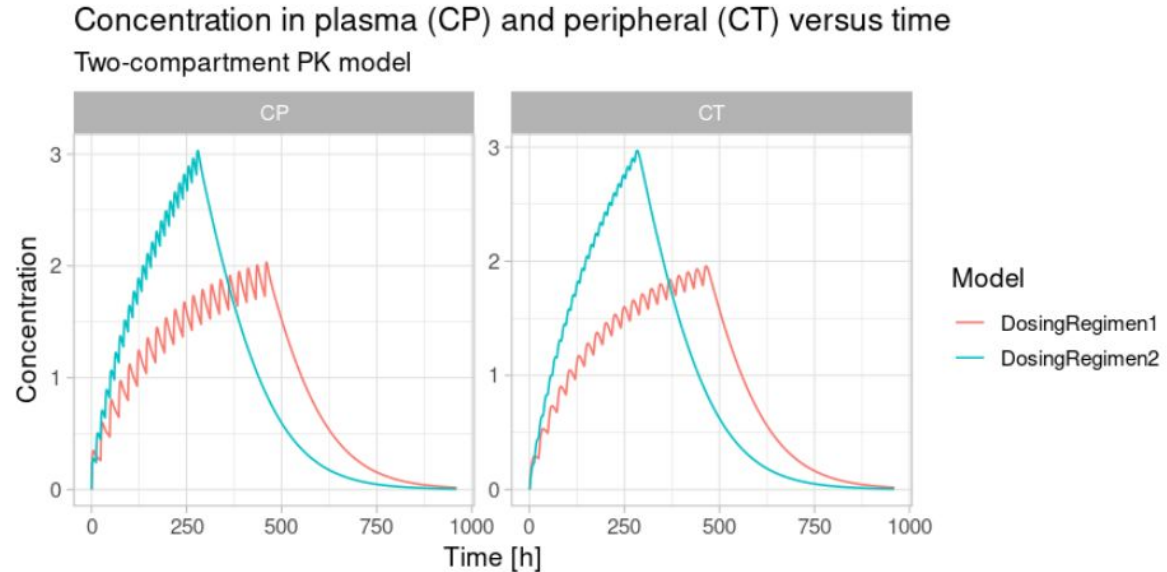
- **VC** (20) Central volume (volume)
- **Q** (2) Inter-compartmental clearance (volume/time)
- **VP** (10) Peripheral volume of distribution (volume)
- **Ka** (K_a , 0.5) Absorption rate constant (1/time)
- **Vmax** (1) Maximum velocity of elimination (mass/time)
- **Km** (3) Michaelis constant for elimination (mass/volume)

Values of the parameters derive from *in vitro* assays (for instance V_{max} and K_m), previous *in vivo* studies, or predictions (for instance with machine learning).

Simulating two dosing regimens

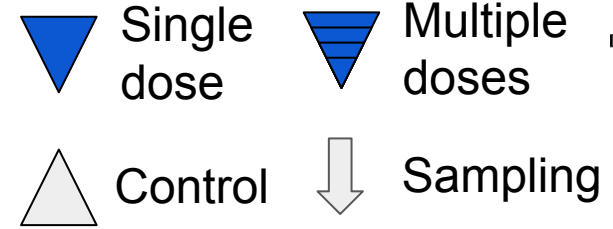
Dosing regimen 1: dosing **8** units per **12** hours, for **24** doses

Dosing regimen 2: dosing **10** units per **24** hours, for **20** doses.

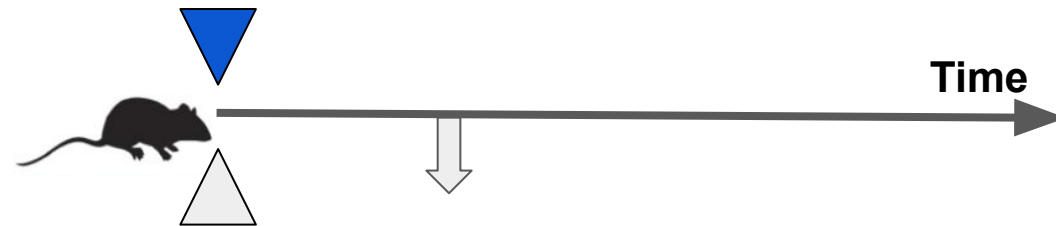


Translational PK/PD Modelling

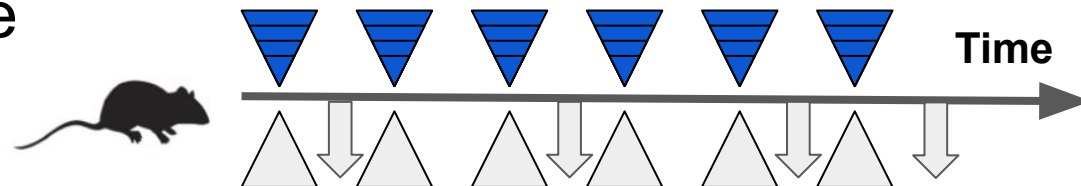
Samples from blood and tissues or organs can be analysed for pharmacokinetics, pharmacodynamics, and dose-exposure-response relationships.



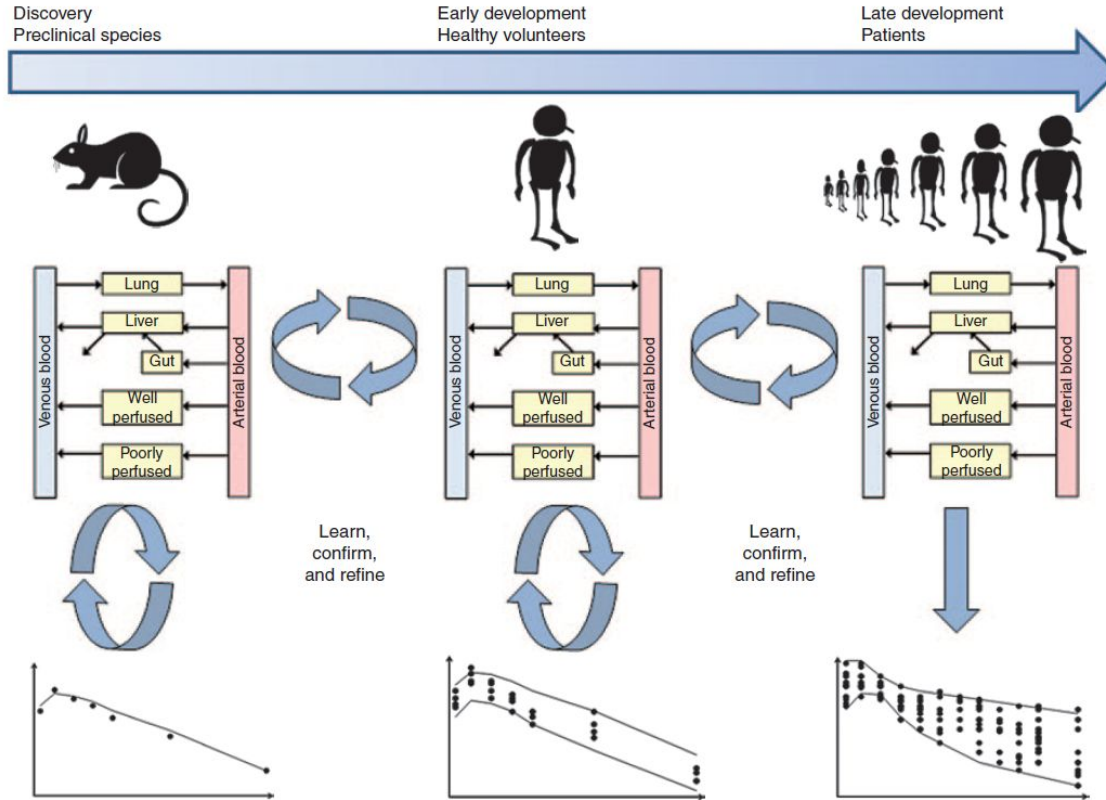
Acute efficacy model



Chronic PK/PD model



Physiologically-based pharmacokinetic modelling (PBPK) is a natural extension of PK modelling



Empirical, stratified, and individualized medicine



Empirical medicine

- Vaccines
- Non-steroid anti-inflammatory drugs (NSAIDs)

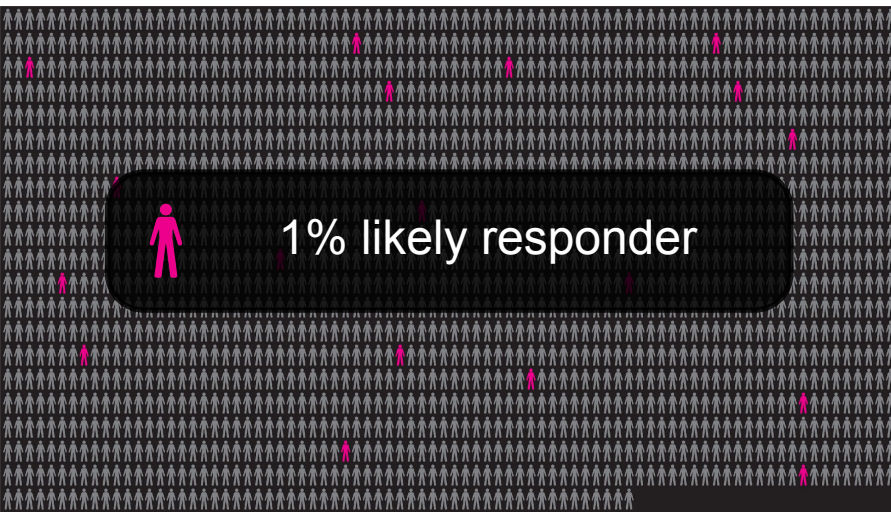
Stratified Medicine

- Vemurafenib (Zelboraf)
- Trastuzumab (Herceptin)

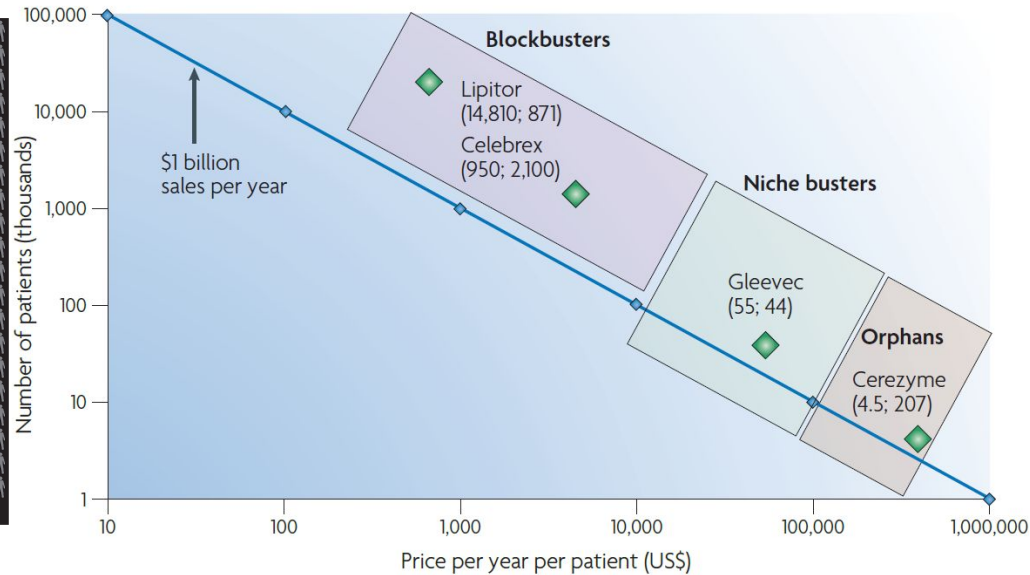
Individualized medicine

- CAR-T therapy

Why stratified medicines are becoming popular?

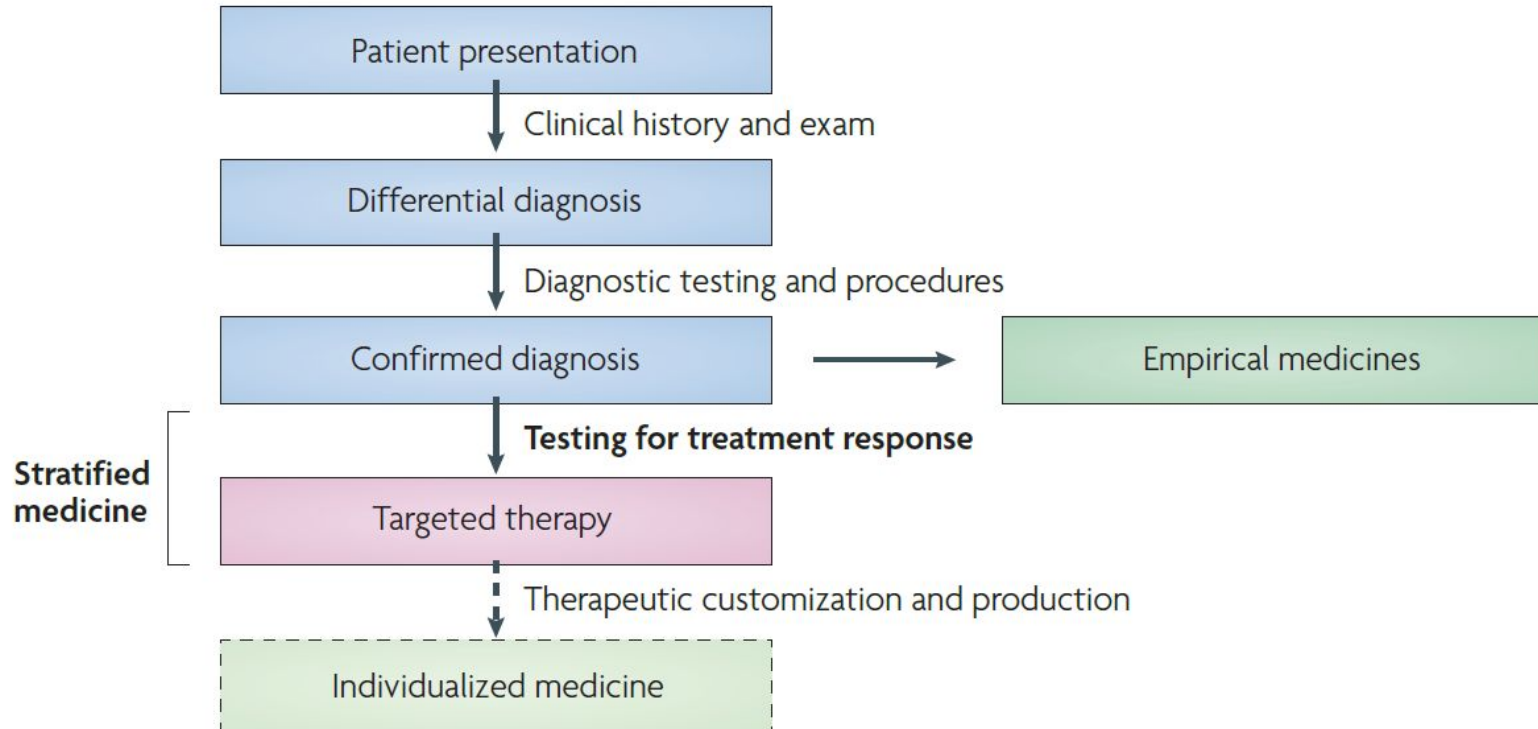


Medical reasons



Commercial reasons

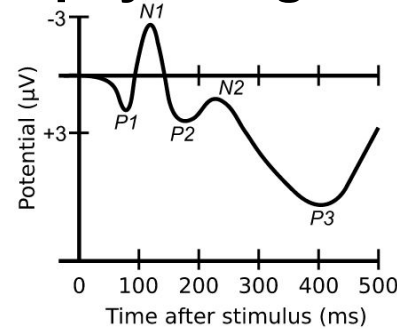
Empirical, stratified, and individualized medicine in the clinical context



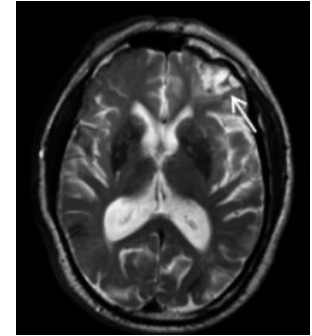
Biomarkers

A objectively measured and evaluated characteristic as an indicator of (1) normal biological process, (2) pathogenic processes, or (3) pharmacological responses to a therapeutic intervention.

Electro-physiological



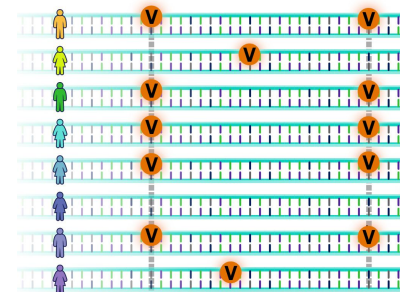
Imaging



Functional

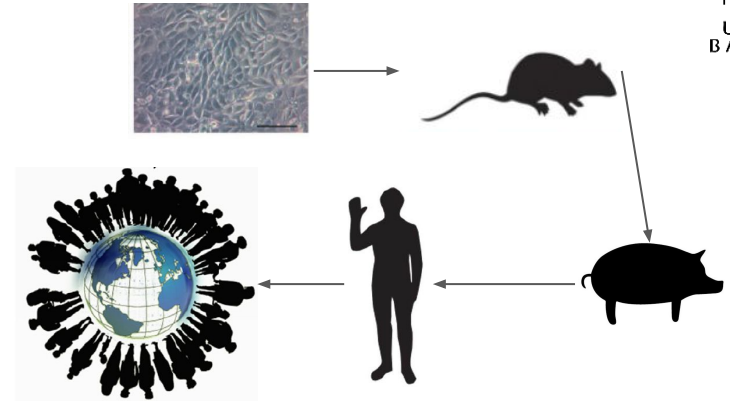


Molecular



Applications of biomarkers

1. Compound optimization and differentiation from competitors in preclinical study
2. Human-dose prediction in translational PK/PD modelling
3. Patient stratification in clinical studies

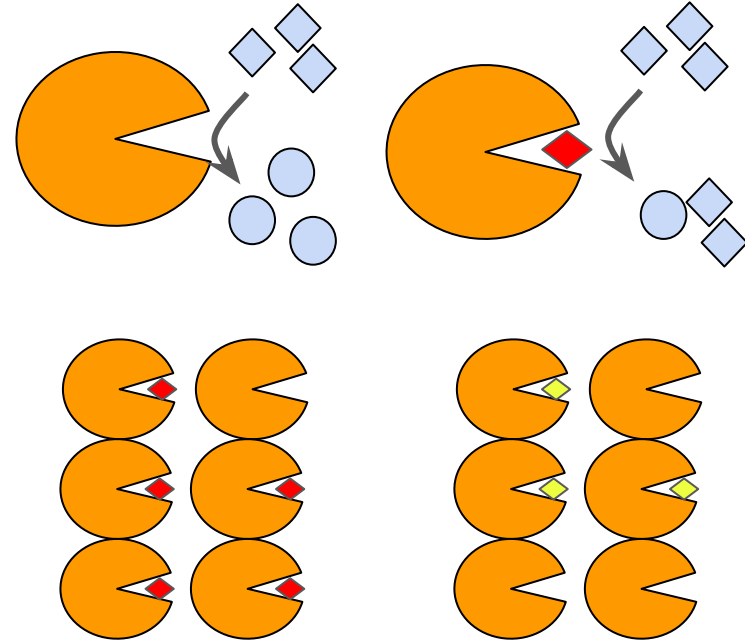


Target Occupancy and Engagement

Target occupancy is the percentage of the protein target occupied by drugs.

Target occupancy affects **target engagement**, which describes the process a drug interacts with its intended protein target in a living system to induce downstream effects. An occupied target is not necessarily engaged: the mode of binding and the physiological context matters.

The mode of binding and the downstream effects are known as the Mechanism of Action (MoA) of the drug.

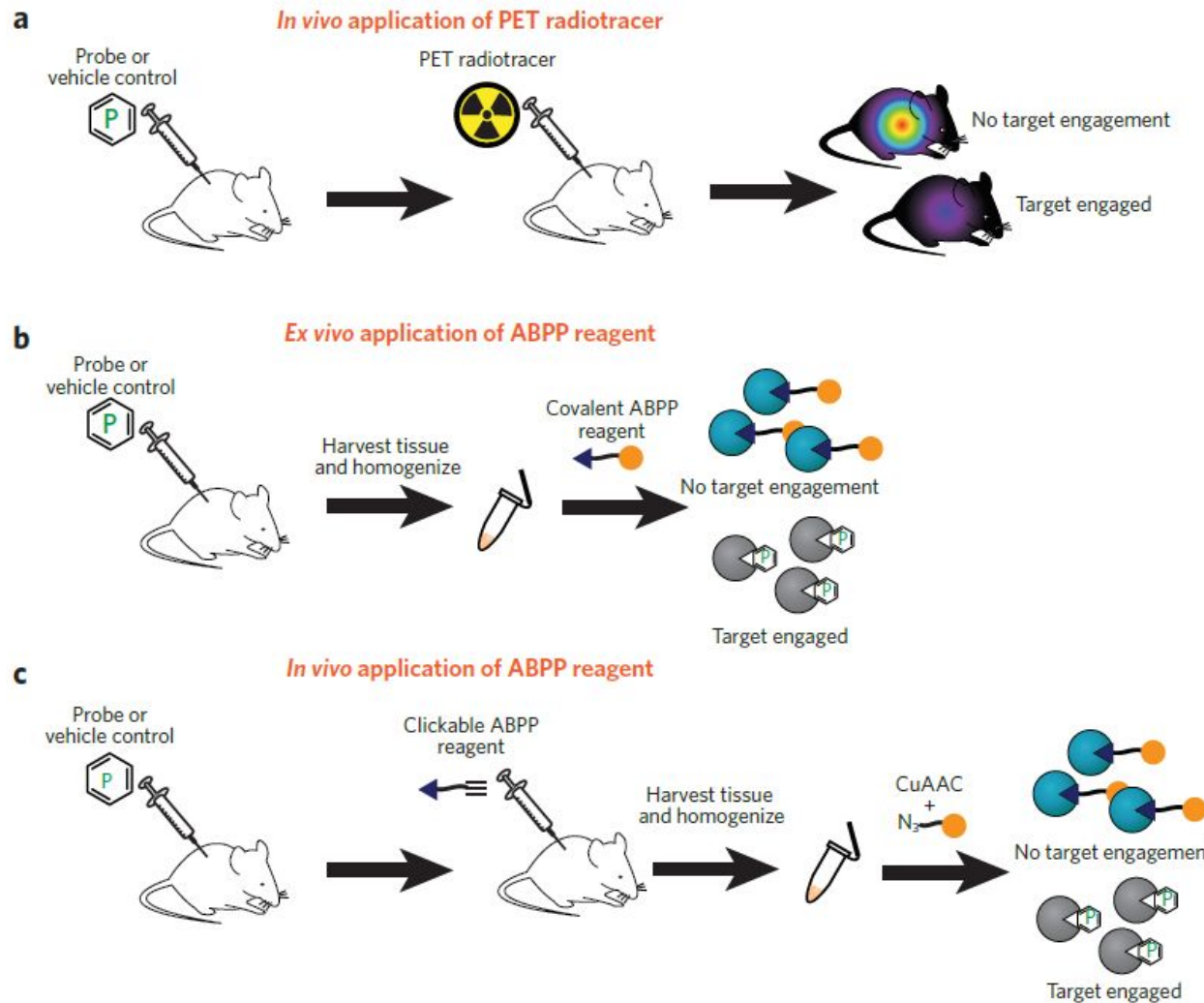


Target occupancy of 83% and 50%, respectively

Target occupancy and engagement profiling in vivo

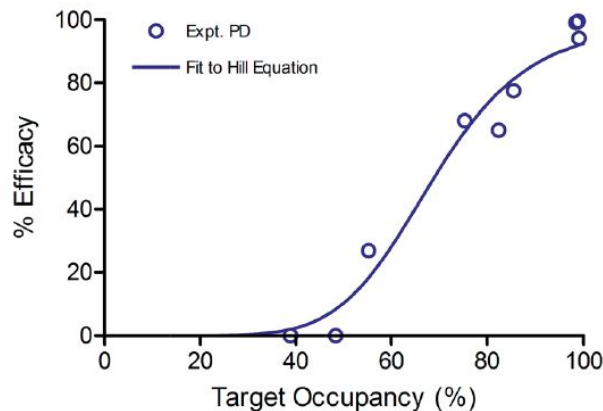
ABPP: Activity-based protein profiling; PET: positron-emission tomography.

Both ABPP reagent and radiotracer binds to the same protein target.

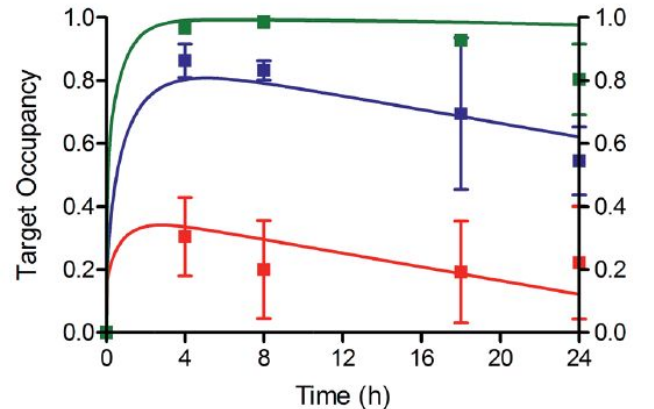


Target occupancy as a biomarker links pharmacokinetics and pharmacodynamics

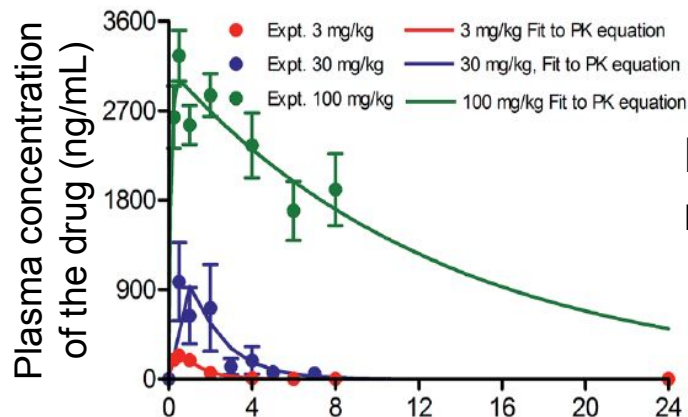
**PD
modelling**



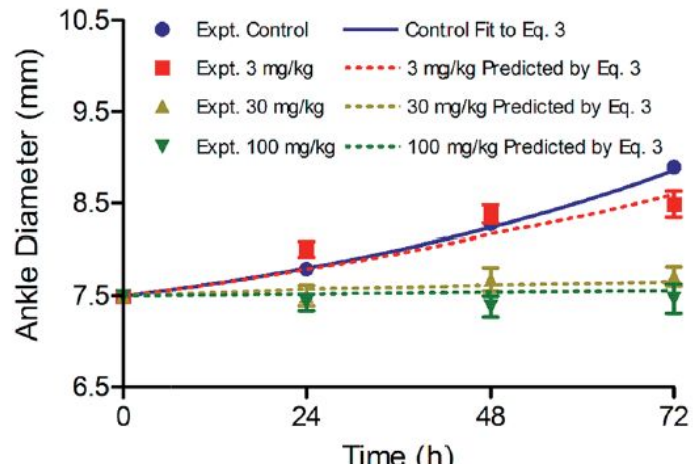
**Biomarker
modelling**



**PK
modelling**



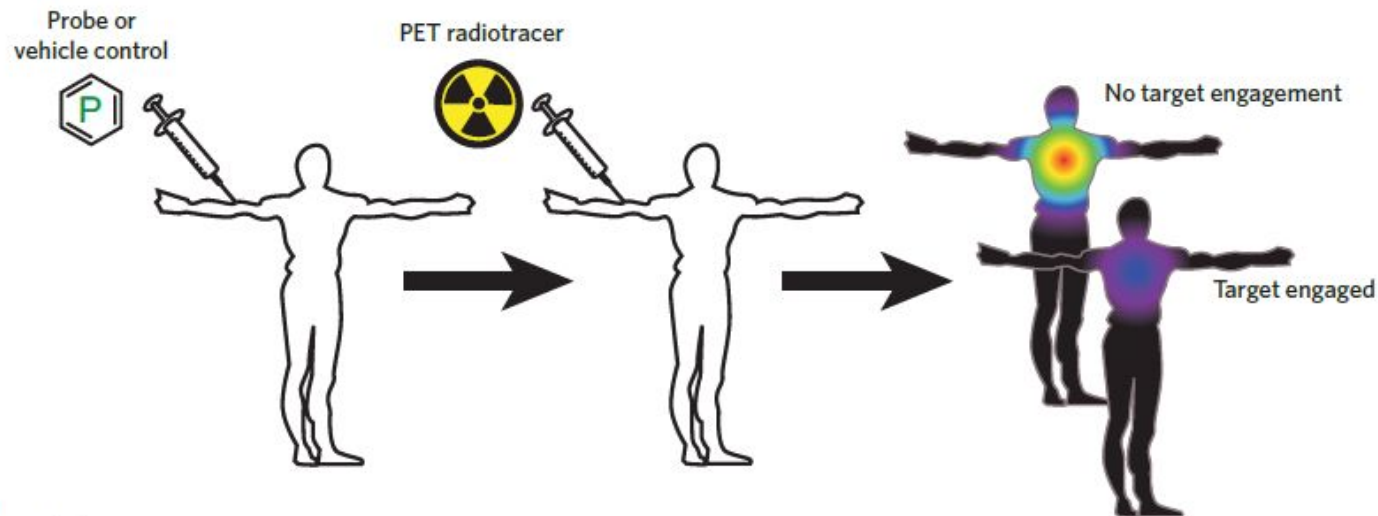
**PK/PD
modelling**



Target occupancy and engagement profiling in human

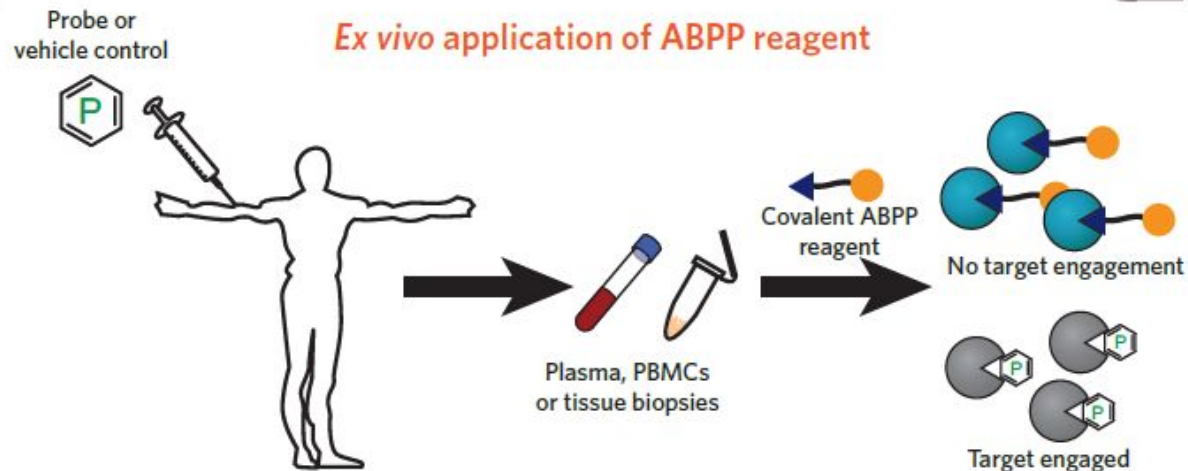
a

In vivo application of PET radiotracer

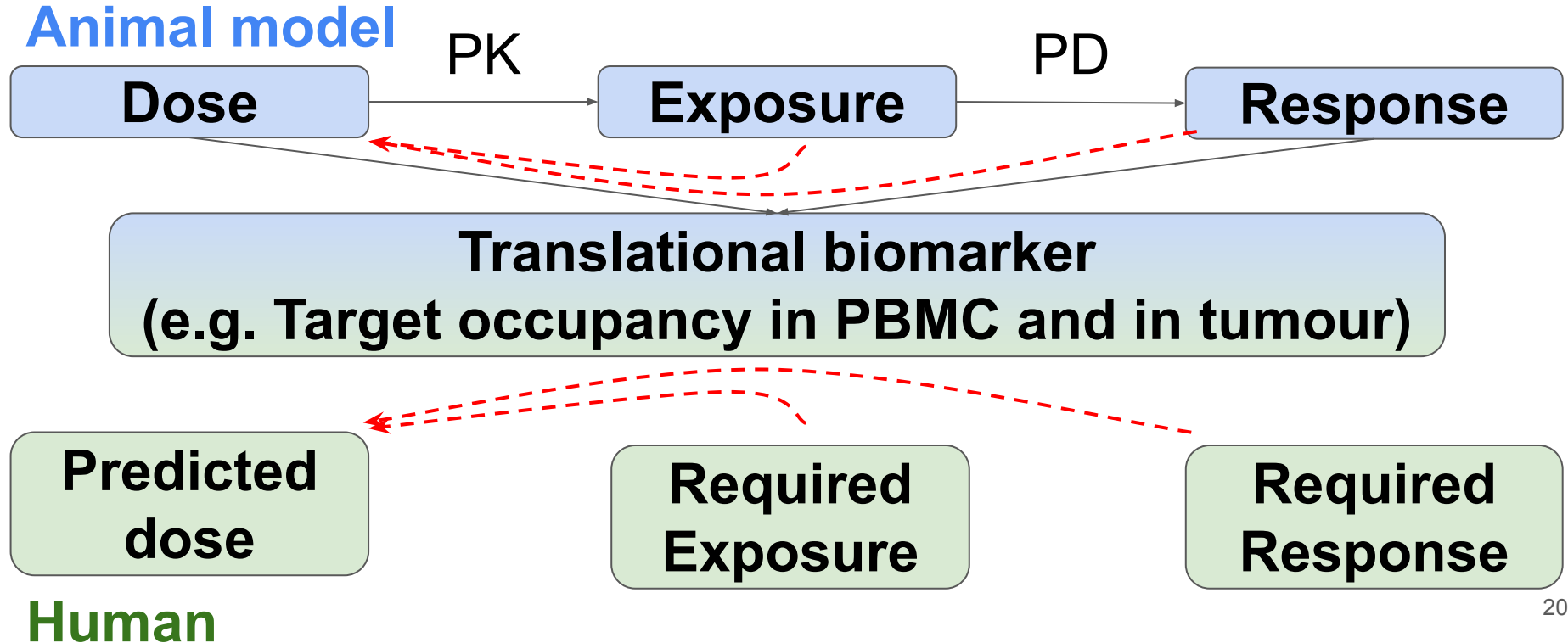


b

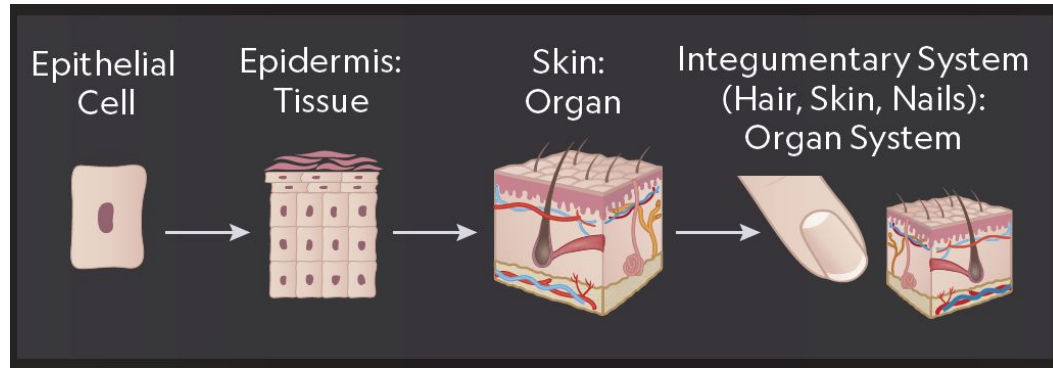
Ex vivo application of ABPP reagent



Use of translational biomarker



Complexity Increases Through a System



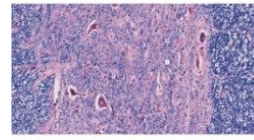
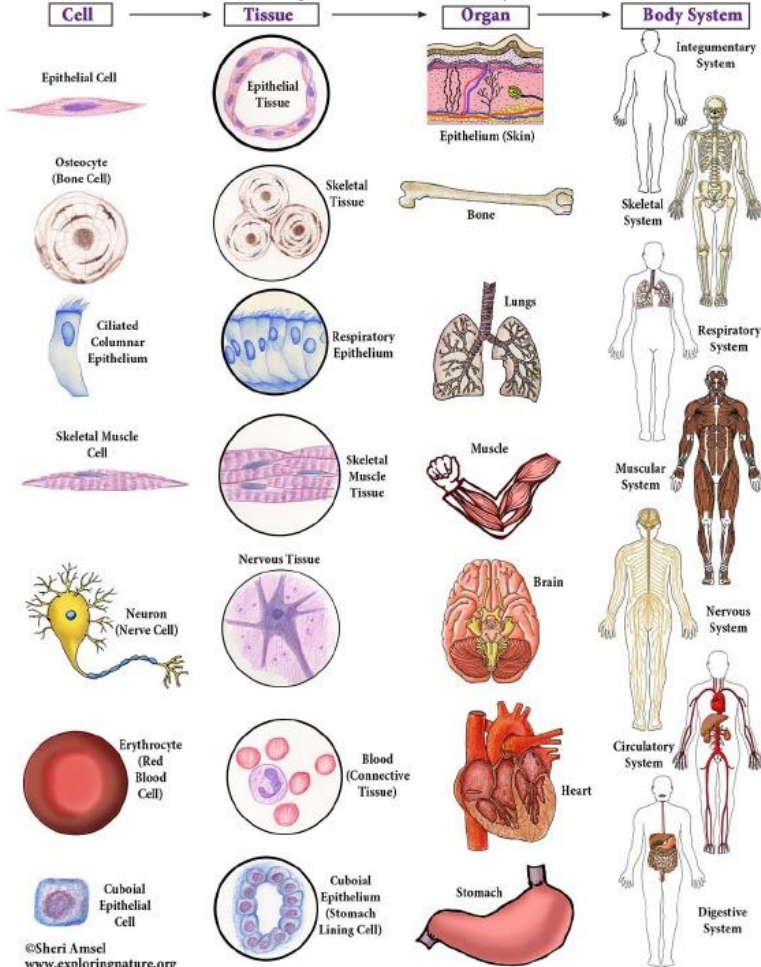
Cells: basic building blocks, variable morphologies and functions

Tissues: groups of specialized cells that communicate and collaborate

Organ: group of tissues to perform specific functions

Organ systems: group of organs and tissues

Organization of the Body

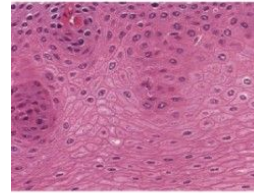


Nervous tissue

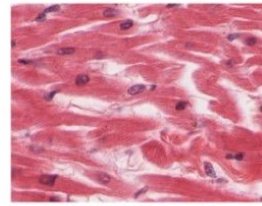
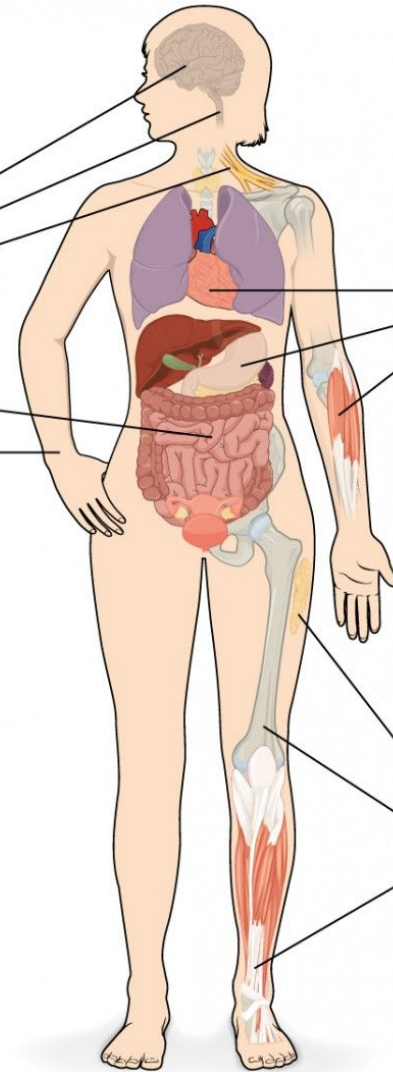
Brain
Spinal cord
Nerves

Epithelial tissue

Lining of GI tract organs
and other hollow organs
Skin surface (epidermis)

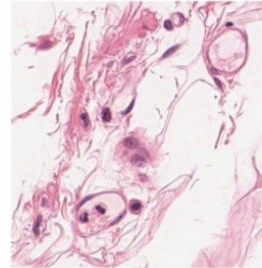


Four major tissue types



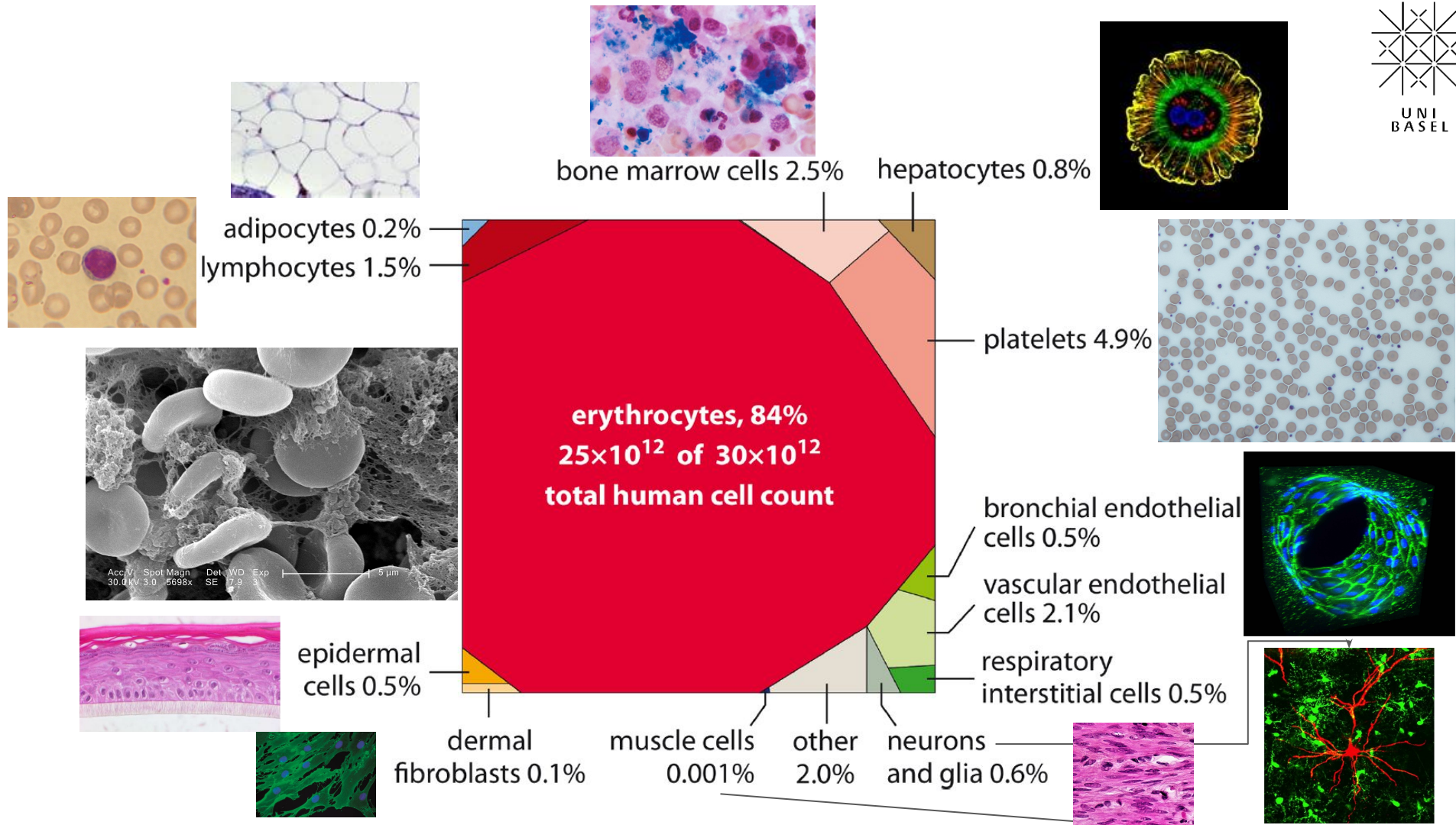
Muscle tissue

Cardiac muscle
Smooth muscle
Skeletal muscle

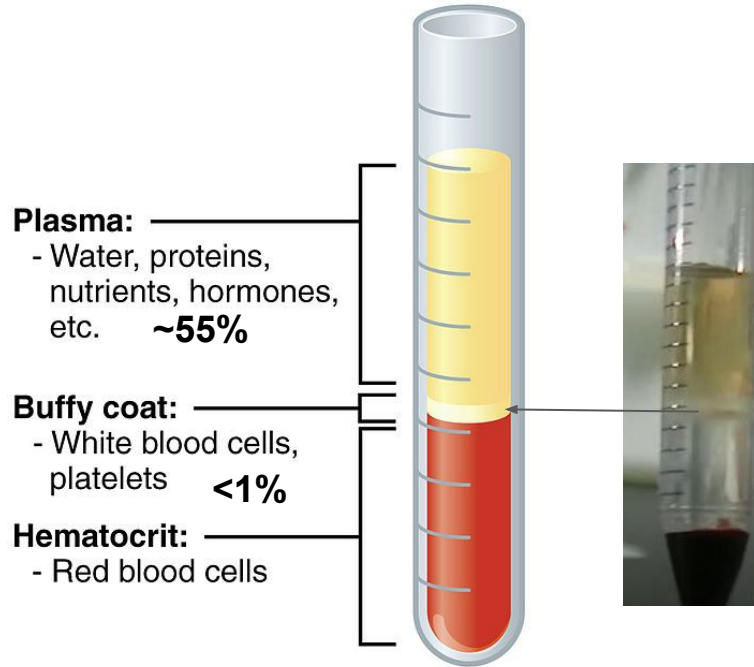


Connective tissue

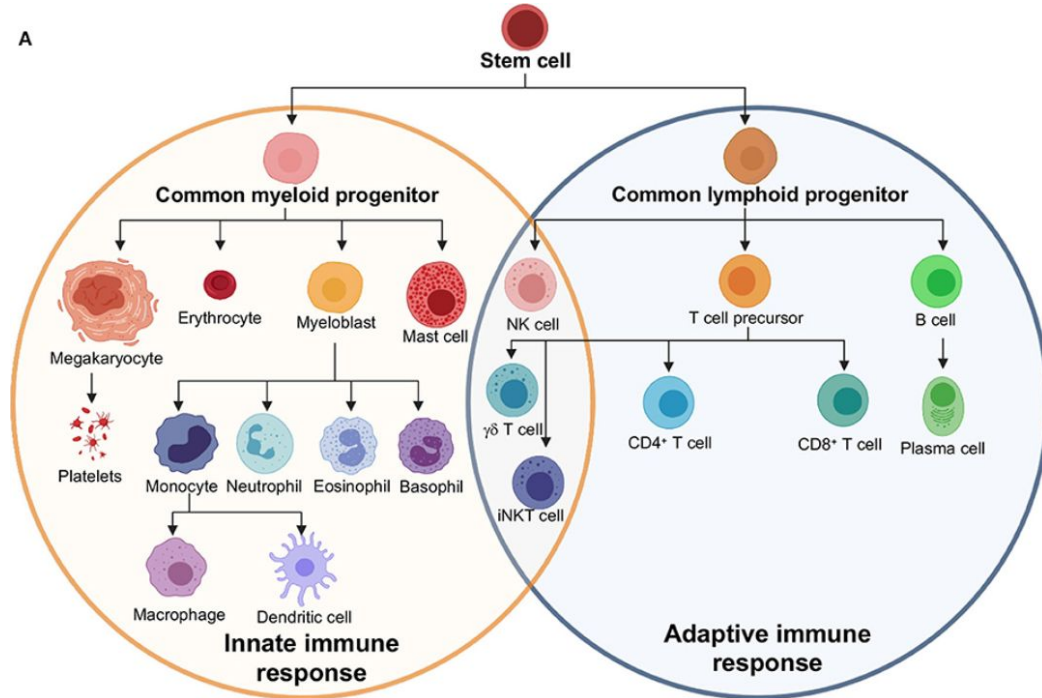
Fat and other soft padding tissue
Bone
Tendon



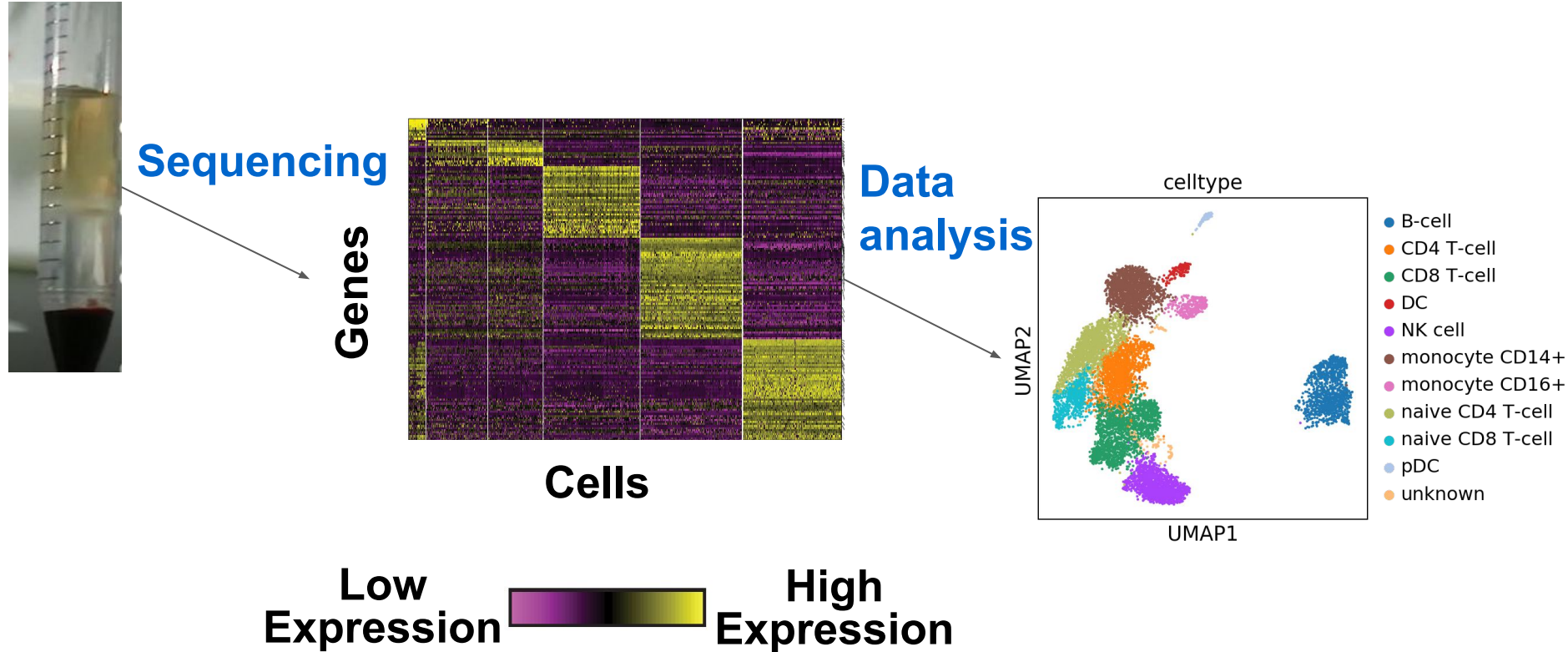
What's in a drop of blood? Ask a doctor or a biologist!



Normal Blood:
♀ 37%–47% hematocrit
♂ 42%–52% hematocrit



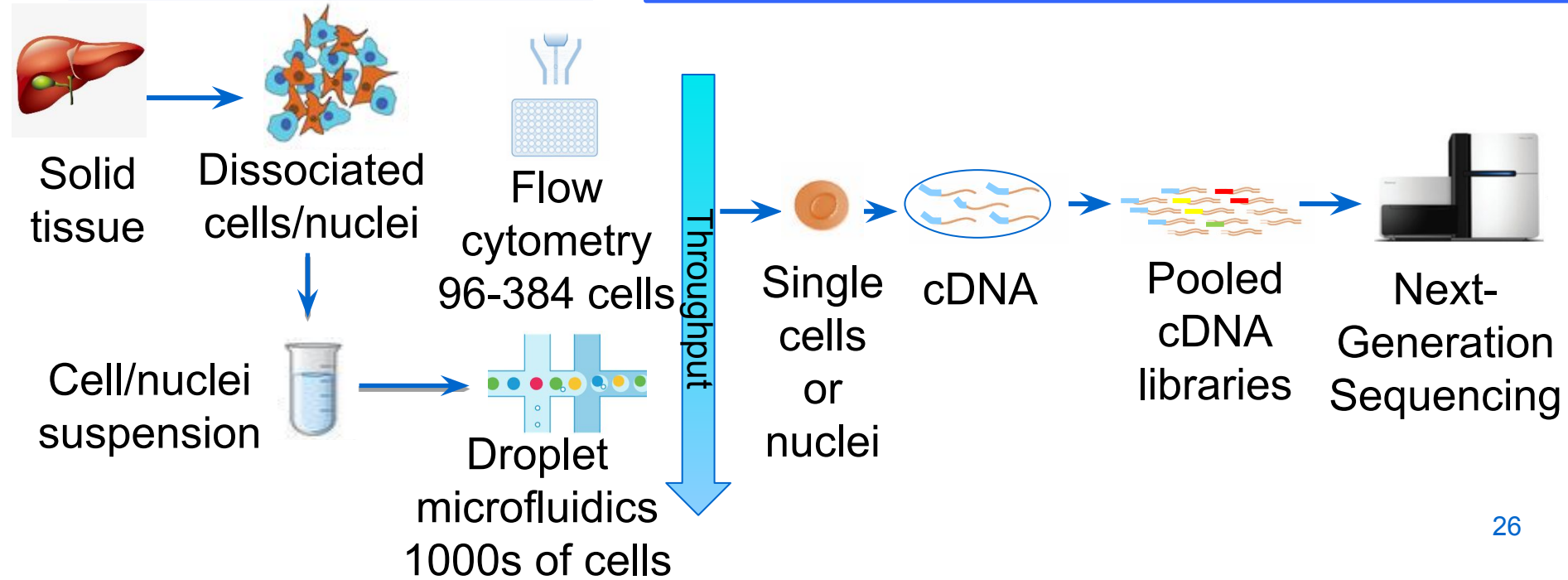
What's in a drop of blood? Count the genes!



Single-cell sequencing (scSeq) workflow

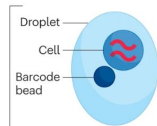
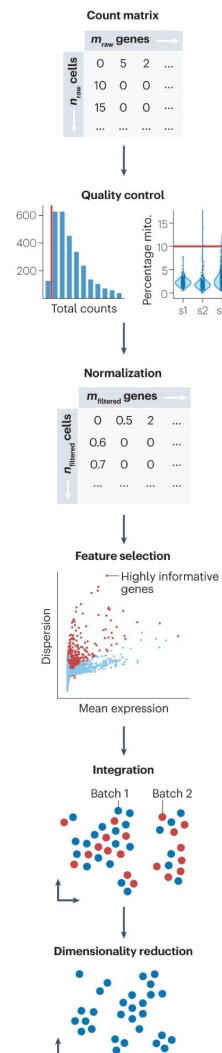
Tissue dissociation

Single cell capture and transcriptome sequencing

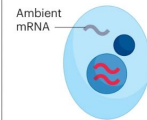


Overview of the computational workflow

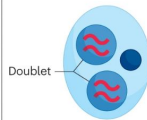
a Preprocessing and visualization



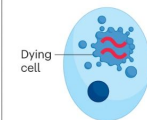
Ambient RNA correction



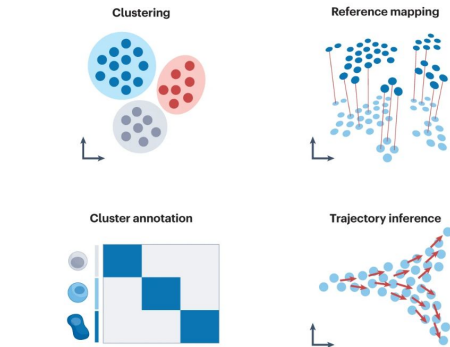
Doublet detection



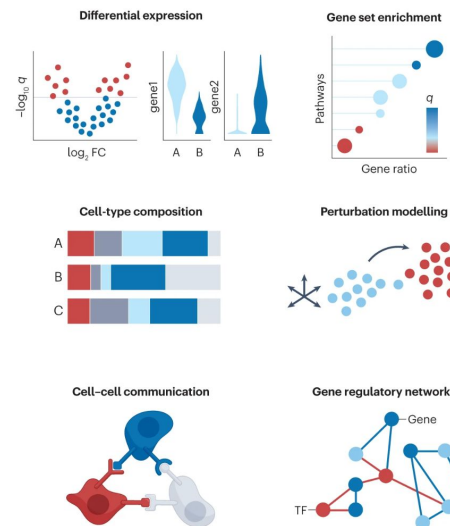
Low-quality cell filtering



b Identifying cellular structure



c Revealing mechanisms

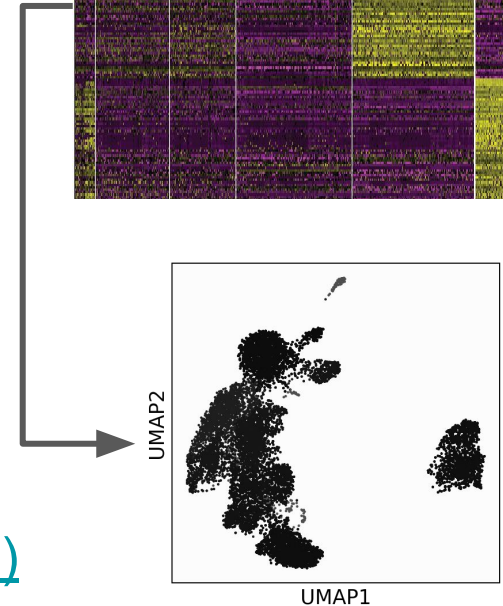
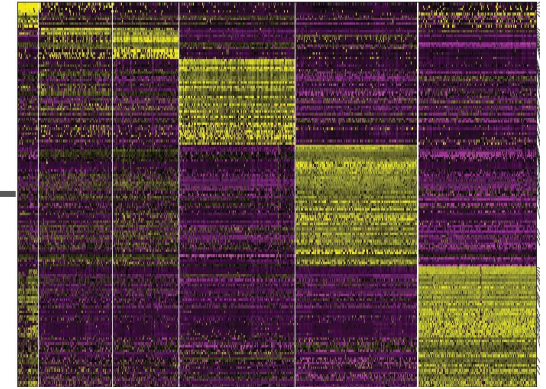
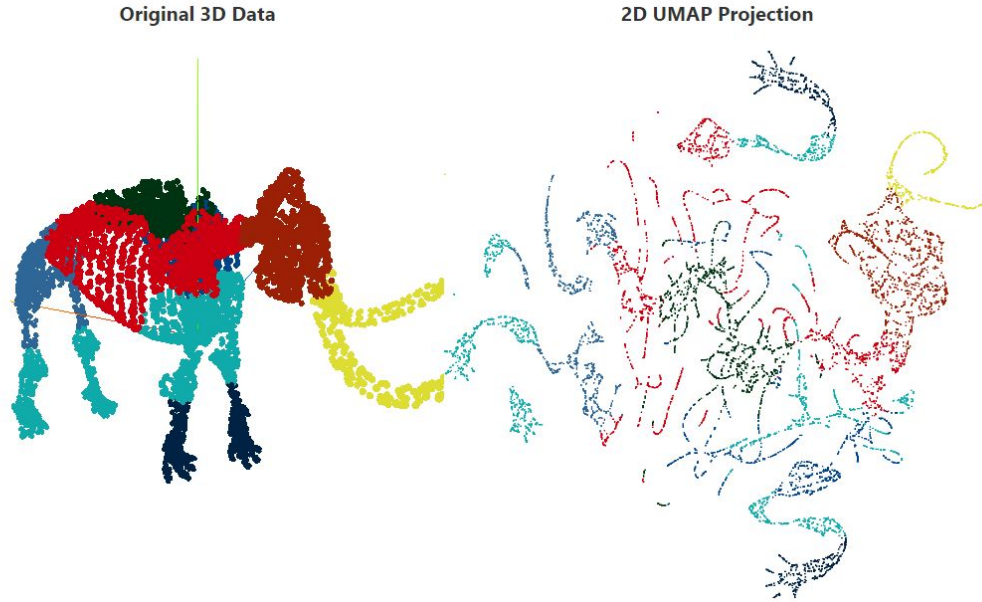


How to represent voxels with pixels?



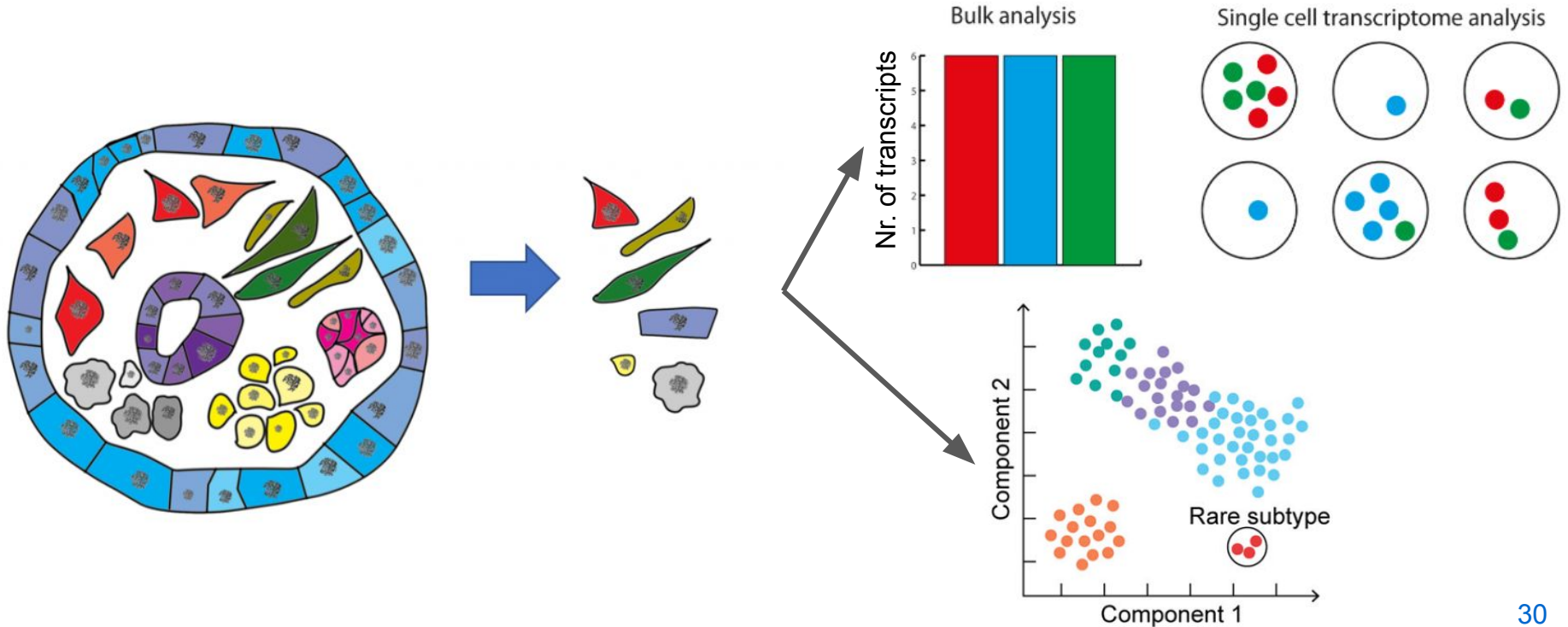
The elephant bull *Tusker* (1992-2023) at Zolli Basel plays with a tree trunk on a post (2022)

Uniform Manifold Approximation and Projection (UMAP) for dimension reduction



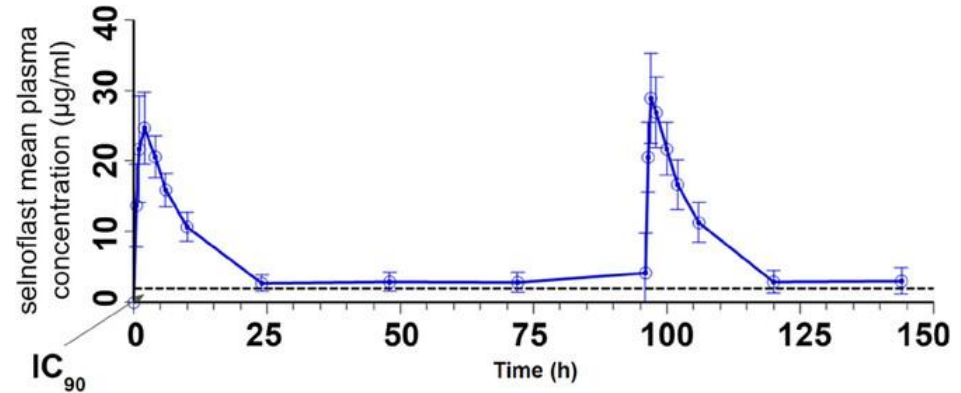
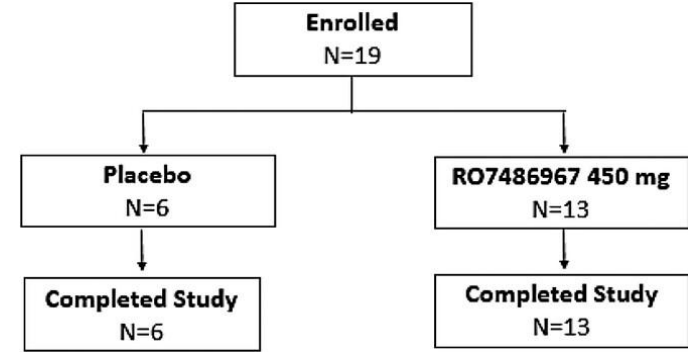
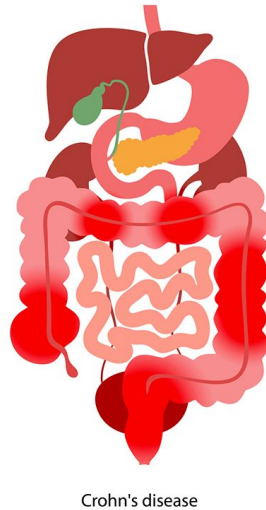
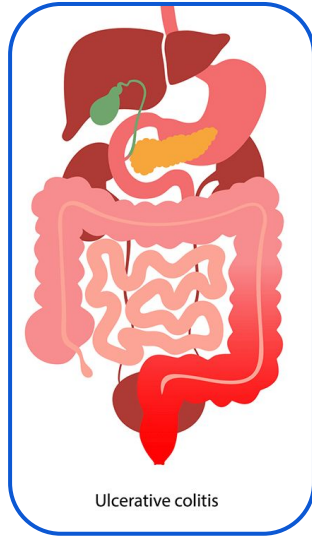
[Understanding UMAP by A. Coenen and A. Pearce](#)
[UMAP by Leland McInnes on SciPy 2018 \(YouTube\)](#)

Single-cell biology benefits both disease understanding and drug discovery



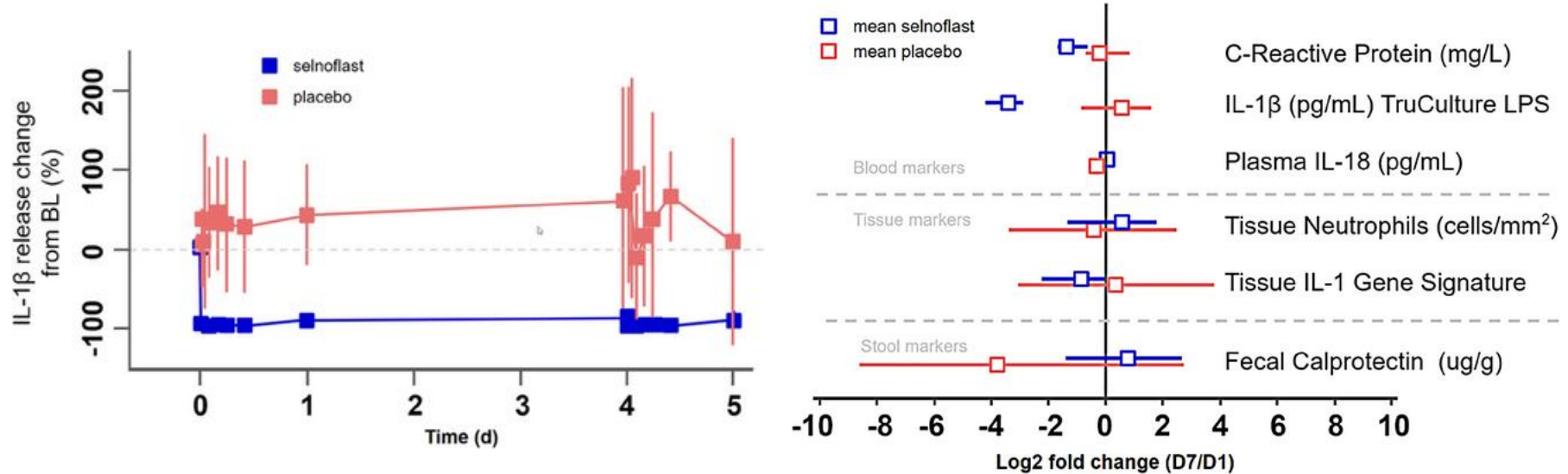
Phase 1b study of Selnoflast in UC

INFLAMMATORY BOWEL DISEASE (IBD)



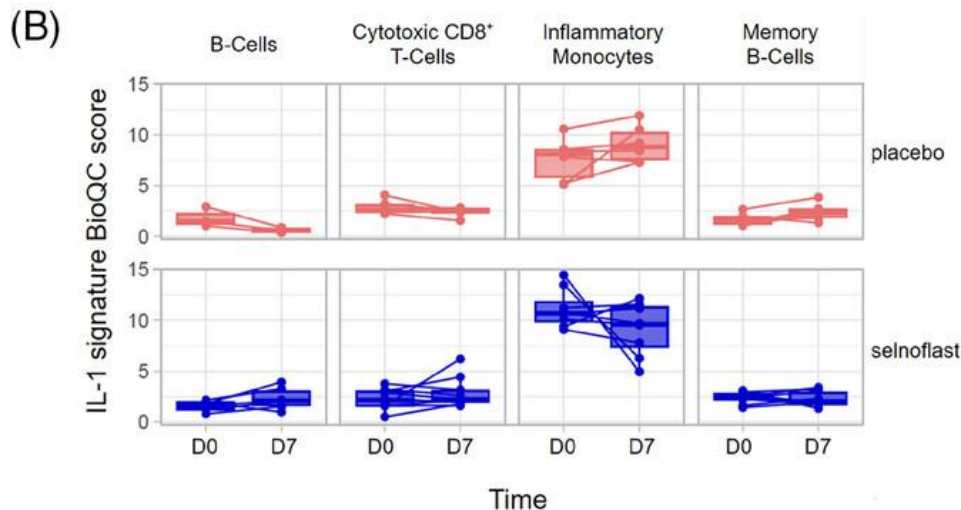
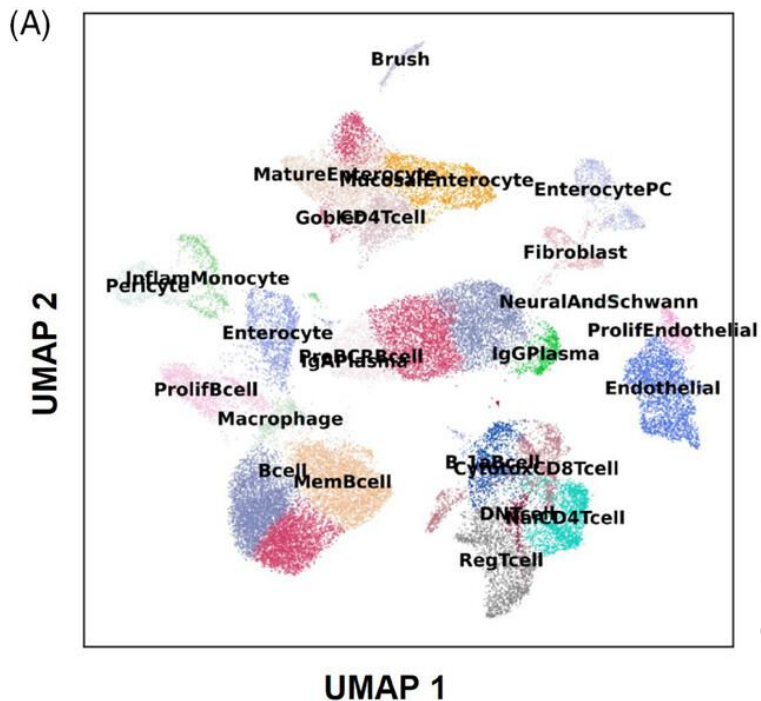
Klughammer, B. et al. A randomized, double-blind phase 1b study evaluating the safety, tolerability, pharmacokinetics and pharmacodynamics of the NLRP3 inhibitor selnoflast in patients with moderate to severe active ulcerative colitis. *Clinical and Translational Medicine* 13, e1471 (2023). IC₉₀ was calculated from *in vitro* studies (2.0 µg/mL or 1.94 µg/g).

Biological assay and omics readout as biomarkers



Selnoflast is a specific antagonist of NLRP3, a protein component of the *inflammasome*. Activation of inflammasome induces interleukin 1 beta (IL-1b), which in turn induces expression of downstream genes (IL-1b gene signatures).

Single-cell RNA-seq revealed that selnoflast failed to induce the changes that we had hoped for



Selnoflast was safe and well-tolerated. Selnoflast 450 mg once a day achieved plasma and tissue exposure predicted to maintain IL-1 β IC₉₀ over the dosing interval. However, PD biomarker results showed no robust differences between treatment arms, suggesting no major therapeutic effects are to be expected in UC.

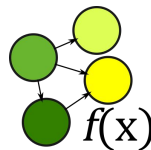
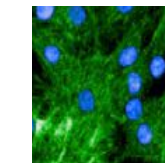
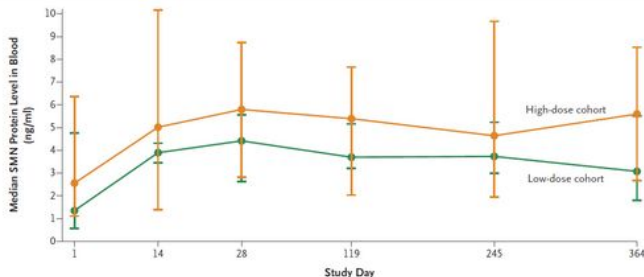
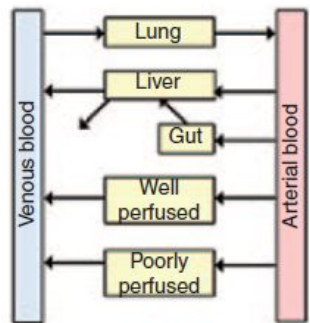
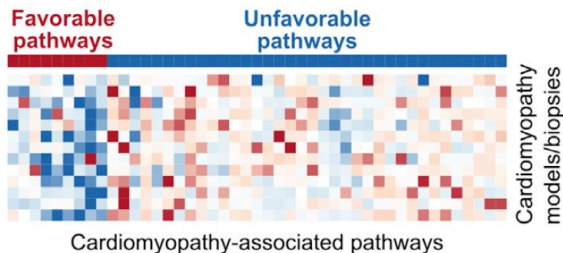
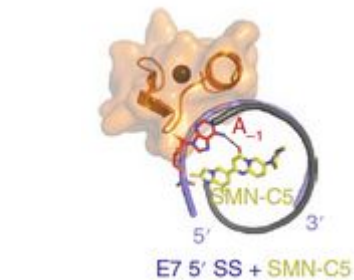
Conclusions

- Biomarkers (1) guide compound optimization and differentiation in preclinical studies, (2) support human dose prediction in translational PK/PD studies, and (3) allow patient stratification in clinical trials;
- Mathematical and computational biology is indispensable for biomarker identification;
- Caveats in biomarker identification calls for integrated mechanistic and statistical modelling to establish causal relations.

Outline of lecture 12

- An example of integrating statistical and mechanistic modelling: Griffiths *et al.*
- Mechanistic modelling of biological systems: from Boolean network to Agent-based modelling
- Causal inference
- Where can we go from here

Drug discovery relies on *in vitro*, *in vivo*, and computational models across scales



Input

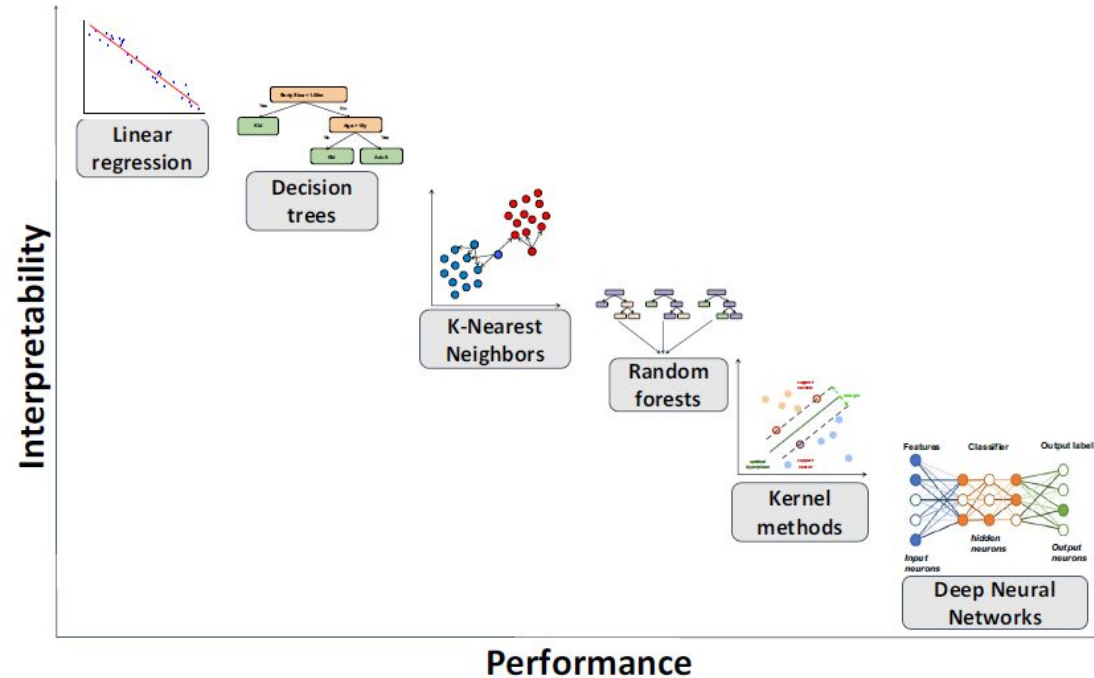


Output

Examples of molecular, omics and cellular, organ and system, and population modelling

Most statistical models predict output without knowing underlying mechanisms

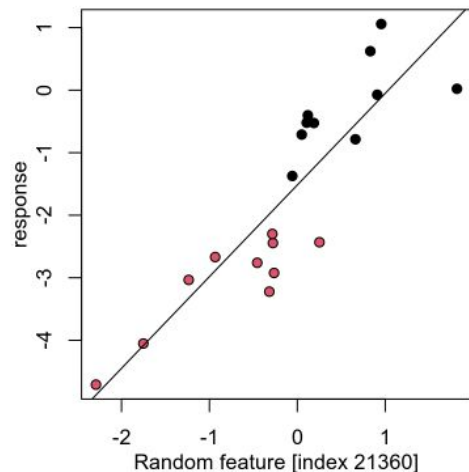
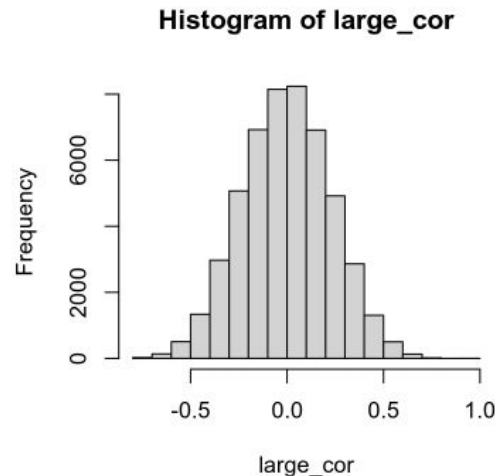
- Non-causal statistical models are useful for hypothesis generation and exploratory analysis.
- Caution is required for *high-dimensional data*, for extrapolation, and for designing interventions.



Simulating the curse of dimensionality

```
set.seed(1887)
patient_group <- gl(2,10)
response <- c(rnorm(10, 0), rnorm(10, -3))
random_features_large <- matrix(rnorm(20*50000), nrow=20)
large_cor <- cor(response, random_features_large, method="spearman")
hist(large_cor)
```

```
largest_cor_ind <- which.max(large_cor)
{
  compactPar()
  plot(random_features_large[, largest_cor_ind],
       response,
       bg=patient_group, pch=21,
       xlab=sprintf("Random feature [index %d]", largest_cor_ind))
  abline(lm(response ~ random_features_large[, largest_cor_ind]))
}
```

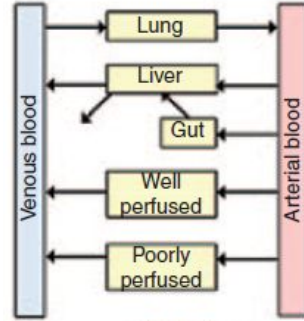


Mechanistic and computational models *explain*

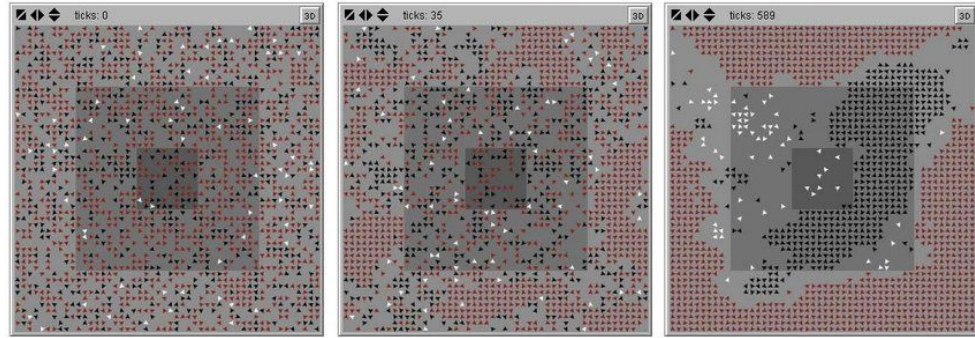
Compartment models

$$\frac{d[LR]}{dt} = k_1[L][R] - k_2[LR]$$

Kinetics of ligand-target interaction

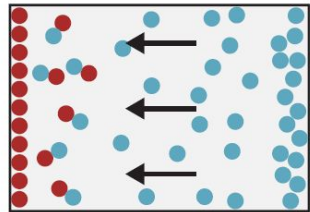


Particle models



Transport models

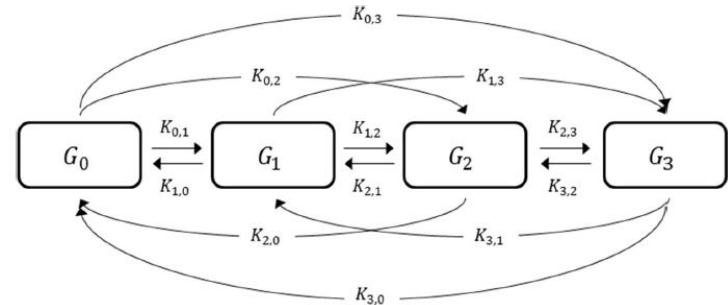
Reaction-Diffusion System



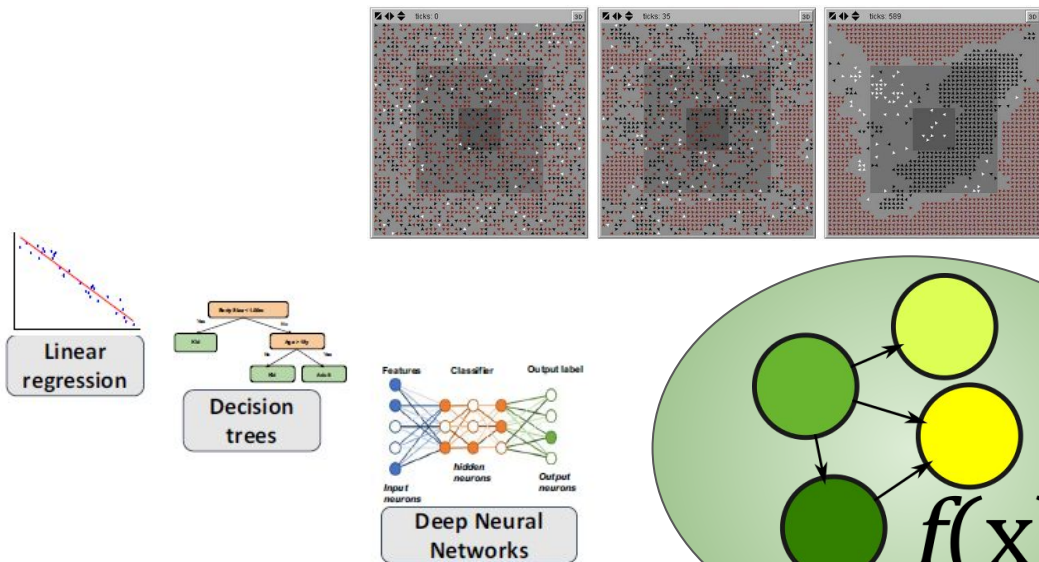
Diffusion

$$\frac{\partial u}{\partial t} = \underbrace{D \frac{\partial^2 u}{\partial x^2}}_{\text{Diffusion}} + \underbrace{ku}_{\text{Binding}}$$

Finite state models

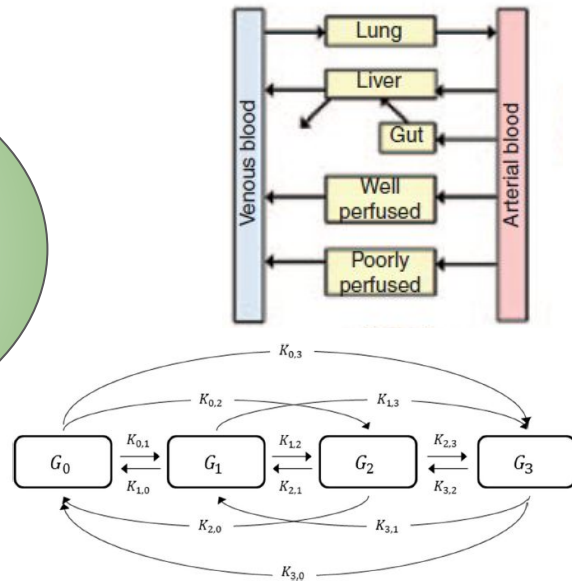


Integration of knowledge, assumptions and data across scales is key for drug discovery



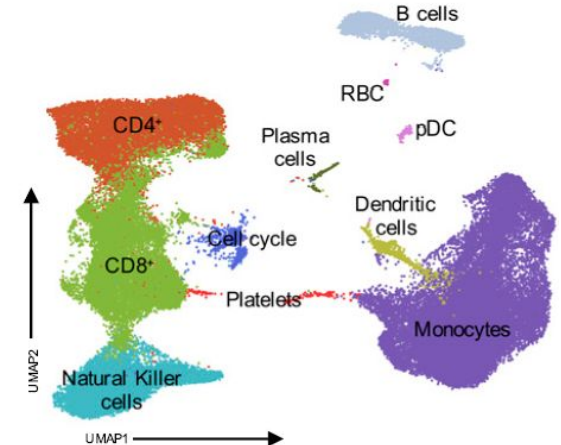
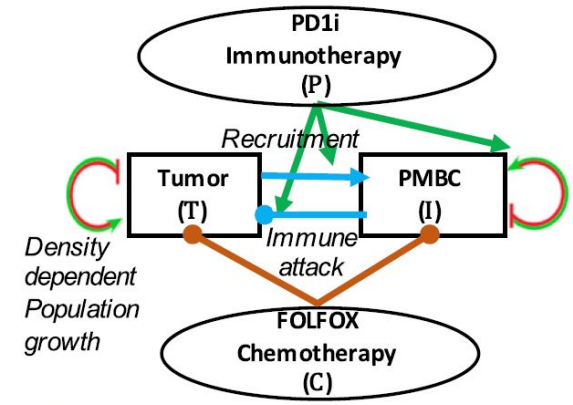
$$\frac{d[LR]}{dt} = k_1[L][R] - k_2[LR]$$

$$\frac{\partial u}{\partial t} = \underbrace{D \frac{\partial^2 u}{\partial x^2}}_{\text{Diffusion}} + \underbrace{ku}_{\text{Binding}}$$



An example of integrated modelling

- Griffiths et al. (PNAS 2020) profiled peripheral immune cell abundance in time series following treatment of Gastrointestinal (GI) tumours with immunotherapy in a small clinical trial.
- The authors used **compartment models** to characterize cell-cell interactions and analysed **single-cell omics data** to reveal immune cell abundance, pathway activity patterns, and differentiation status.



How this study enriches our knowledge

- **Facts**

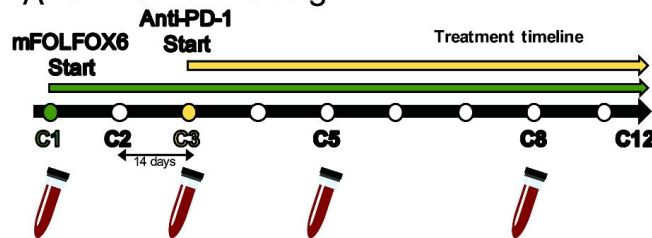
- Clinical response to immune checkpoint inhibitors varies substantially.
- Possible contributing factors correlate only weakly with patient response, including (1) tumor cell mutational load and antigen production, (2) immune-cell infiltration and signalling status, (3) Cross-talk between tumour and immune cells.
- It is challenging to obtain tumour tissue samples.

- **Questions:** Can circulating immune cells serve as a surrogate measurement of a tumour's interaction with the host immune cells and reflect response to therapy early in the course of treatment?
- **Conclusions:** It is possible to predict patient response with the evolution of peripheral immune cell abundance and signalling over time, as well as how immune cell interact with the tumor.

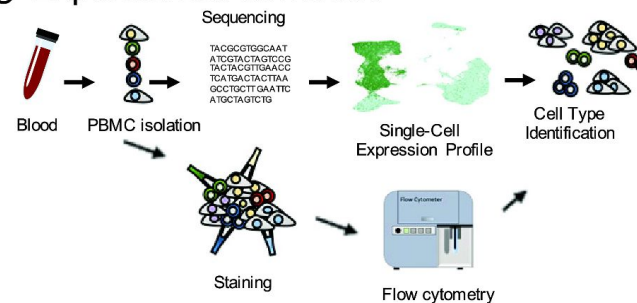
Design of the clinical trial

- mFOLFOX6 (modified FOLFOX6): a chemotherapy regiment.
- Patient response was assessed by RECIST (Response Evaluation Criteria in Solid Tumors) 1.1 guidelines, using computer tomography (CT).

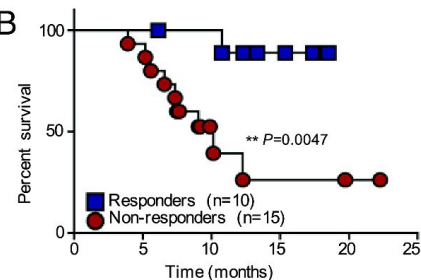
A Clinical trial design



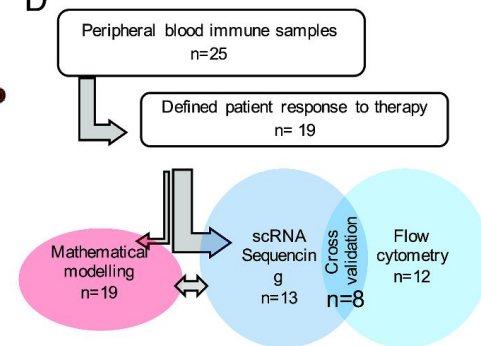
C Experimental flowchart



B



D



Mathematical modelling of tumour-immune cell interactions

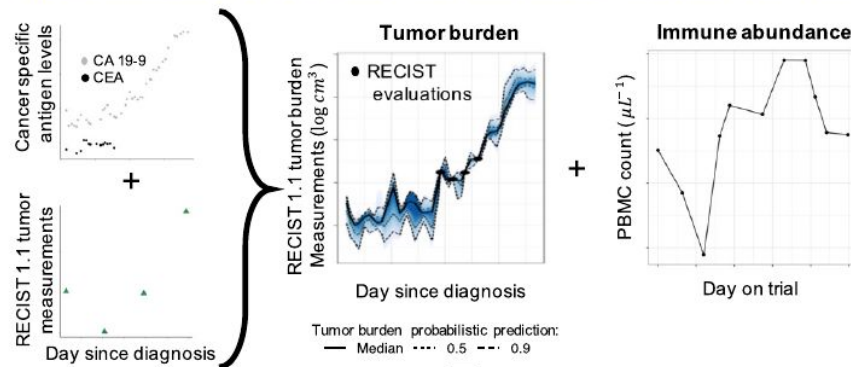
Model inputs (all in time series):

- Tumour burden, inferred by combining antigen values and RECIST evaluation with a *Gaussian process* latent variable model.
- Abundance of PBMCs

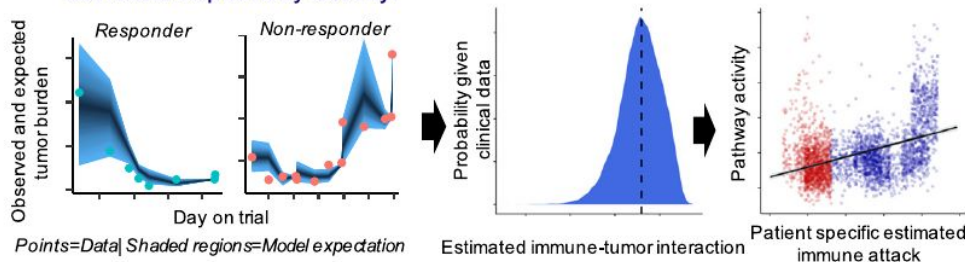
Model output: estimated ability of immune cells to kill tumour cells

E Mathematical model flowchart: tumor-immune cell interactions

i) Construct time course of tumor and immune abundance for each patient:



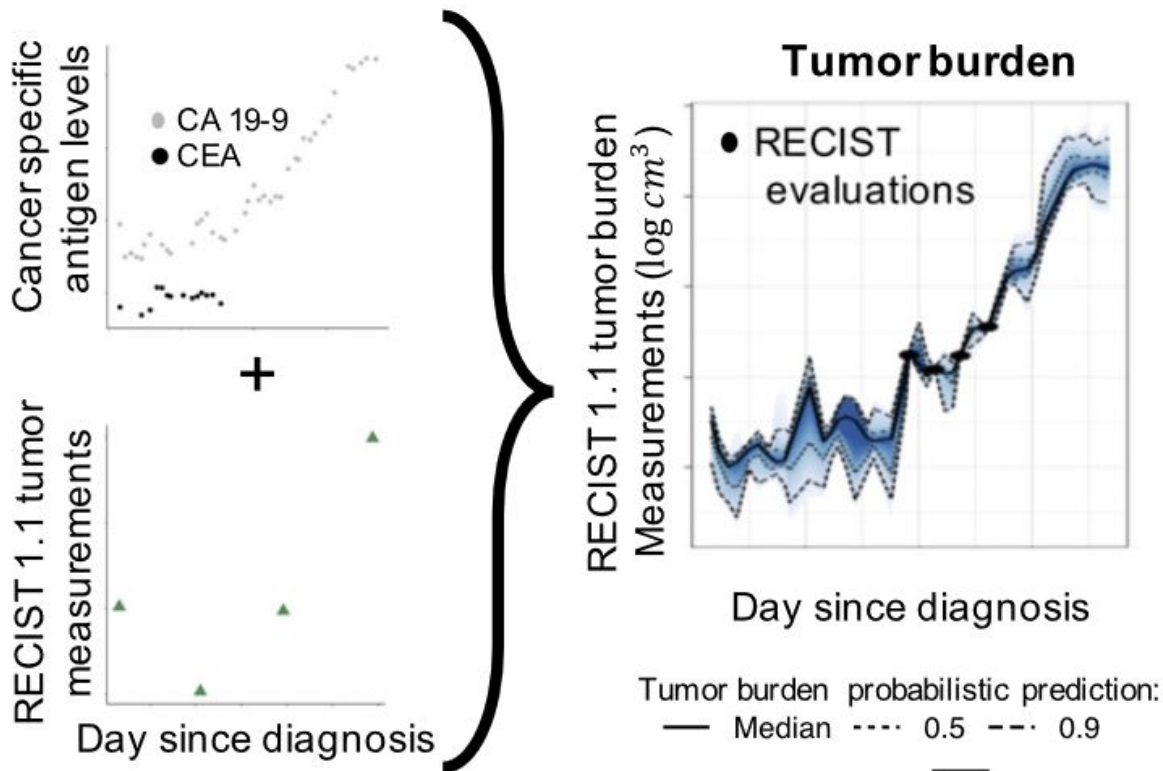
ii) Model how strongly immune cells interact and attack tumor and correlate to pathway activity:



Modelling time-series data with Gaussian Process

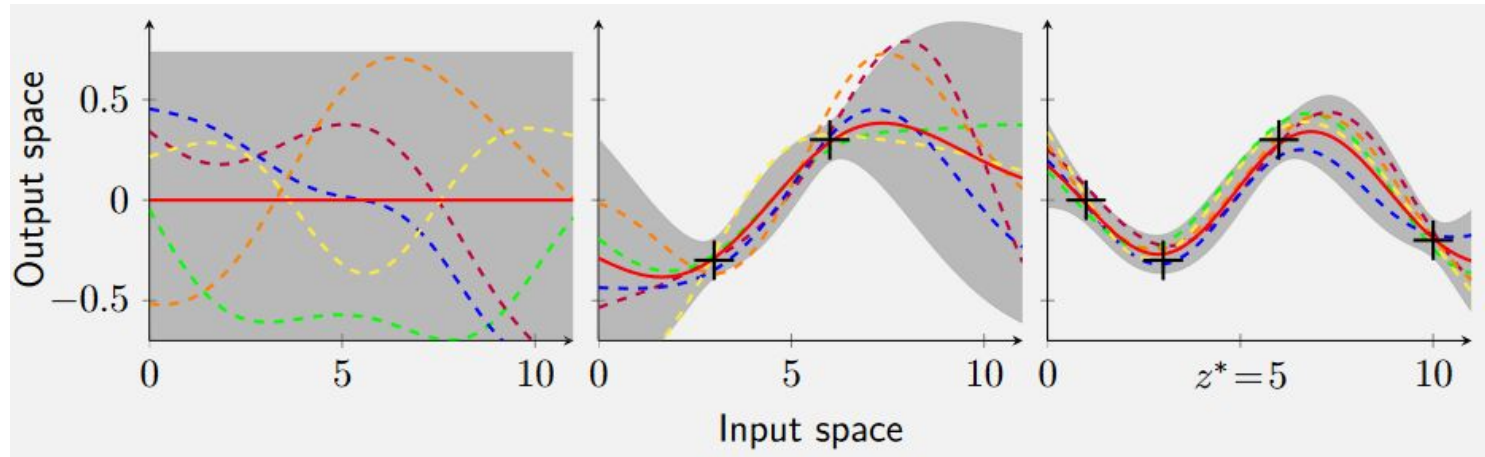
Challenge: tumor growth assessment (RECIST) involves much labour and costly.

Solution: assuming that the data is generated by a Gaussian process, and that cancer-specific antigens is correlated with the tumor growth, we can 'impute' the missing data with other biomarker data with *Gaussian Process*.



Intuitions about Gaussian Process

1. Observe y values as if they are generated from a multivariate Gaussian distribution with *indefinite* dimensions, and time-dependent correlations.
2. We can infer the autocorrelation (i.e. kernel function) by data: the more data, the better we can infer.



The Lotka-Volterra model of predator-prey relationships

- The Lotka-Volterra equations modelling predator-prey relationships.

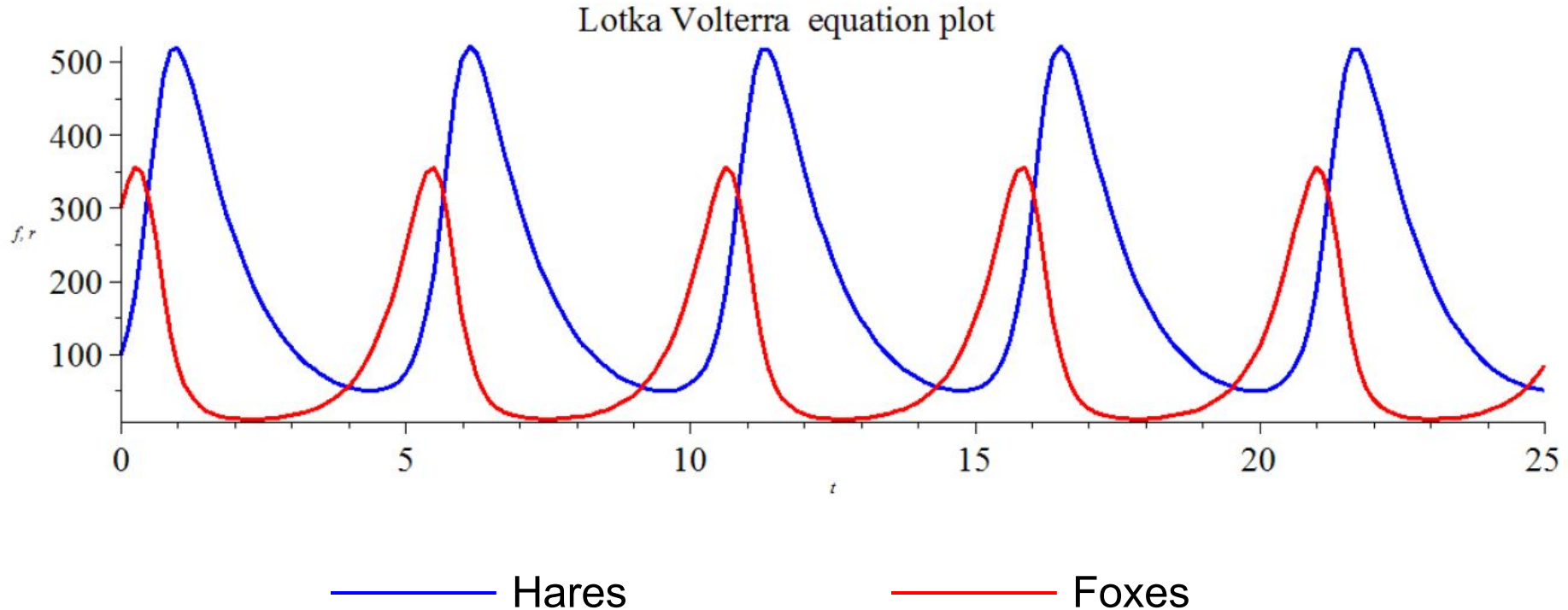
$$\frac{dx}{dt} = \alpha x - \beta xy, \quad (1)$$

$$\frac{dy}{dt} = -\gamma y + \delta xy, \quad (2)$$

where

- x is the number of prey (*e.g.* rabbits),
- y is the number of predator (*e.g.* foxes),
- $\frac{dx}{dt}$ and $\frac{dy}{dt}$ represent growth rates of the two populations,
- t represents time,
- α , β , γ , and δ are real parameters specifying the interaction of the two species.

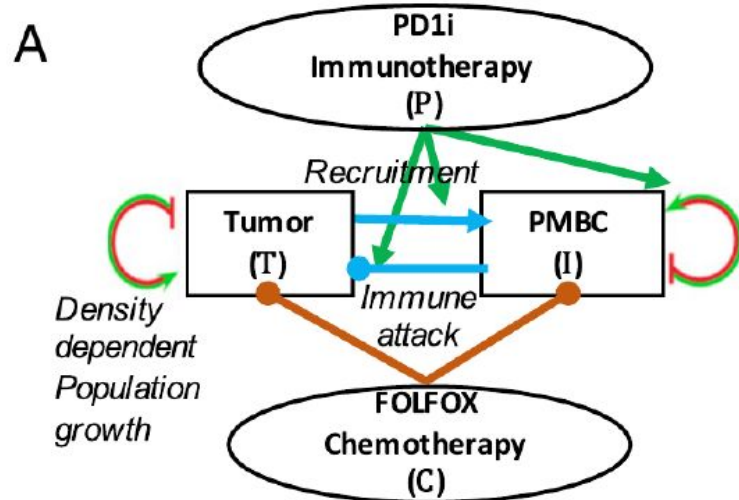
The Lotka-Volterra equations, visualized



Modelling of interactions between tumour and immune cells

Modelling assumptions:

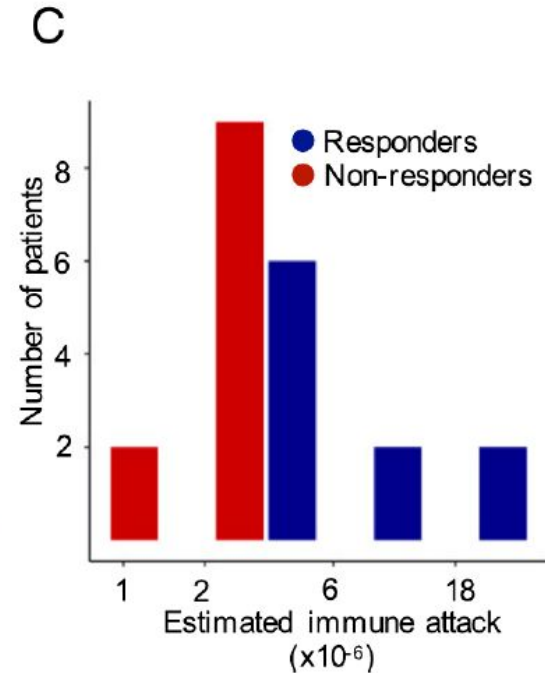
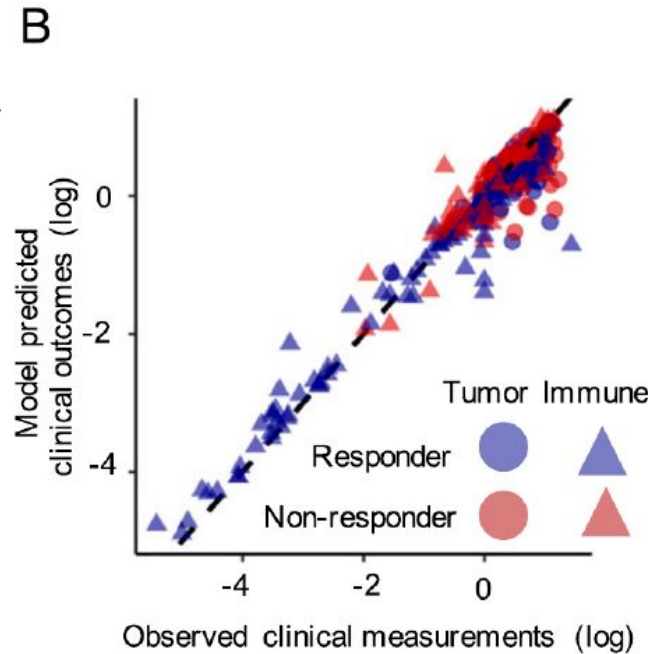
- Tumor cells are attacked by immune cells
- Tumor cells recruit immune cells
- Chemotherapy kills both tumour and immune cells
- Anti-PD1 inhibitor immunotherapy impacts immune proliferation, recruitment, and cytotoxic tumor activity.



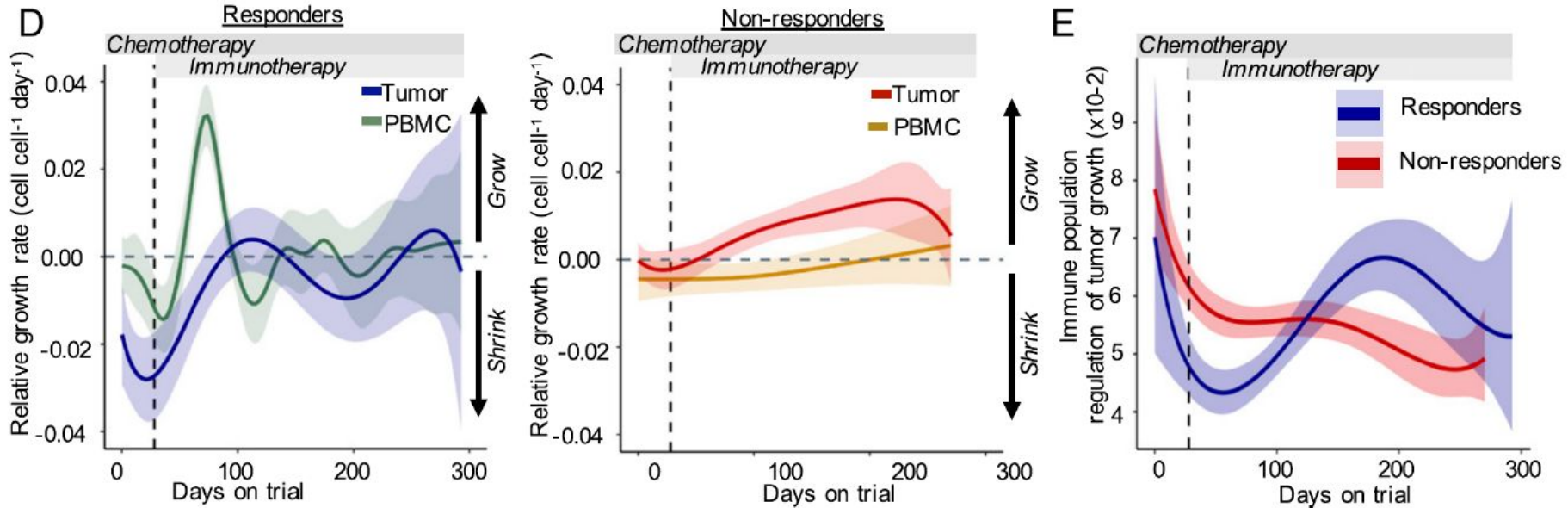
$$\begin{aligned}
 RGR_T &= \frac{1}{T} \frac{dT}{dt} = \underbrace{r_T(1 - \gamma_T T)}_{\text{Tumor growth}} - \underbrace{(\alpha + \beta_\phi P)I}_{\text{Immune attack}} - \sum_i \underbrace{\vec{\mu}_T[i] C_i}_{\text{Chemotherapy}} \\
 RGR_I &= \frac{1}{I} \frac{dI}{dt} = \underbrace{(r_I + \beta_r P)(1 - \gamma_I I)}_{\text{Immune growth}} + \underbrace{(\lambda + \beta_\lambda P)T}_{\text{Recruitment}} - \sum_i \underbrace{\vec{\mu}_I[i] C_i}_{\text{Chemotherapy}}
 \end{aligned}$$

Model prediction and performance

- The strength of immune-tumor interaction is estimated by statistically fitting the growth rate of immune cells and tumor size to model predictions.
- Changes in tumor burden and immune cell abundance are described by data fitting, using a Bayesian hierarchical model.



Profiles of relative growth rates differ between responders and non-responders

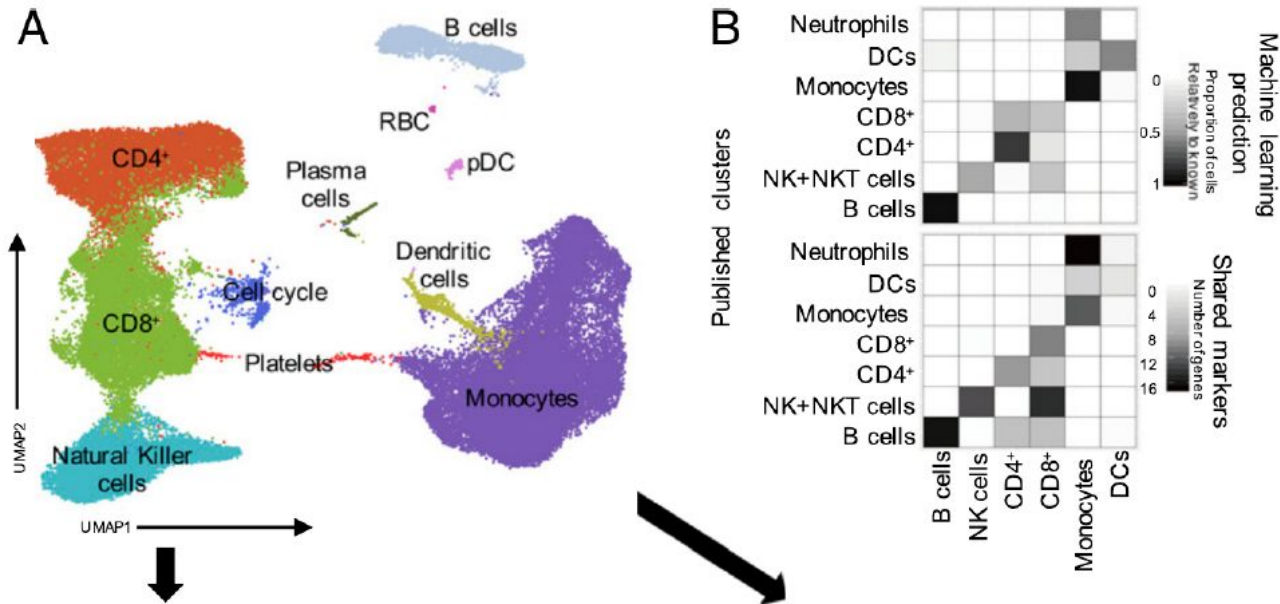


- Neither tumor nor PBMC responds to chemotherapy in non-responders.
- Responders have lower PBMC abundance in general at baseline.

Immune cell population identified by scRNAseq

PBMCs were analysed at three time points:

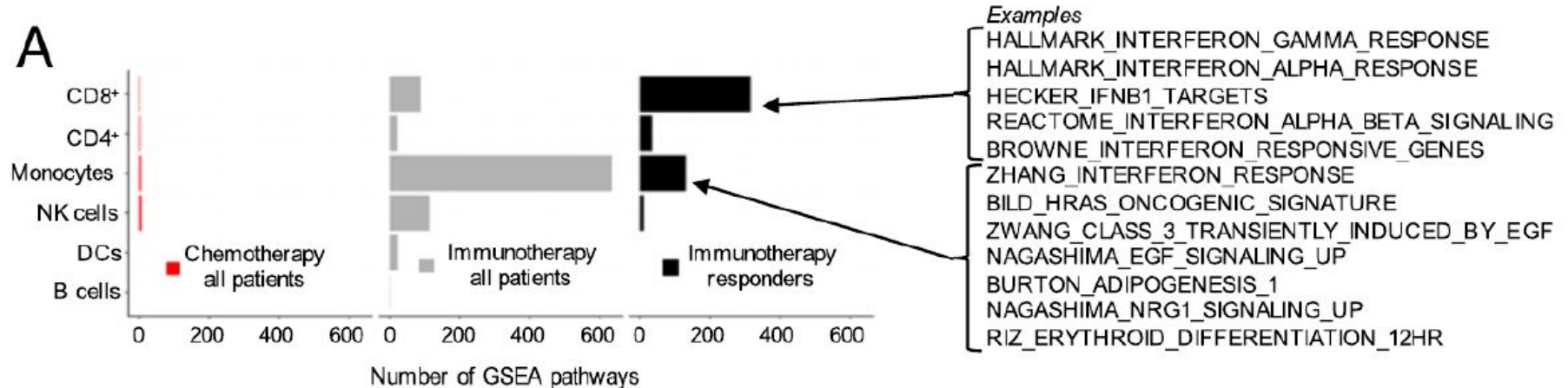
1. Cycle 1 (C1): baseline before treatment;
2. Cycle 3 (C3): chemotherapy alone;
3. Cycle 5 (C5): chemotherapy + anti-PD-1.



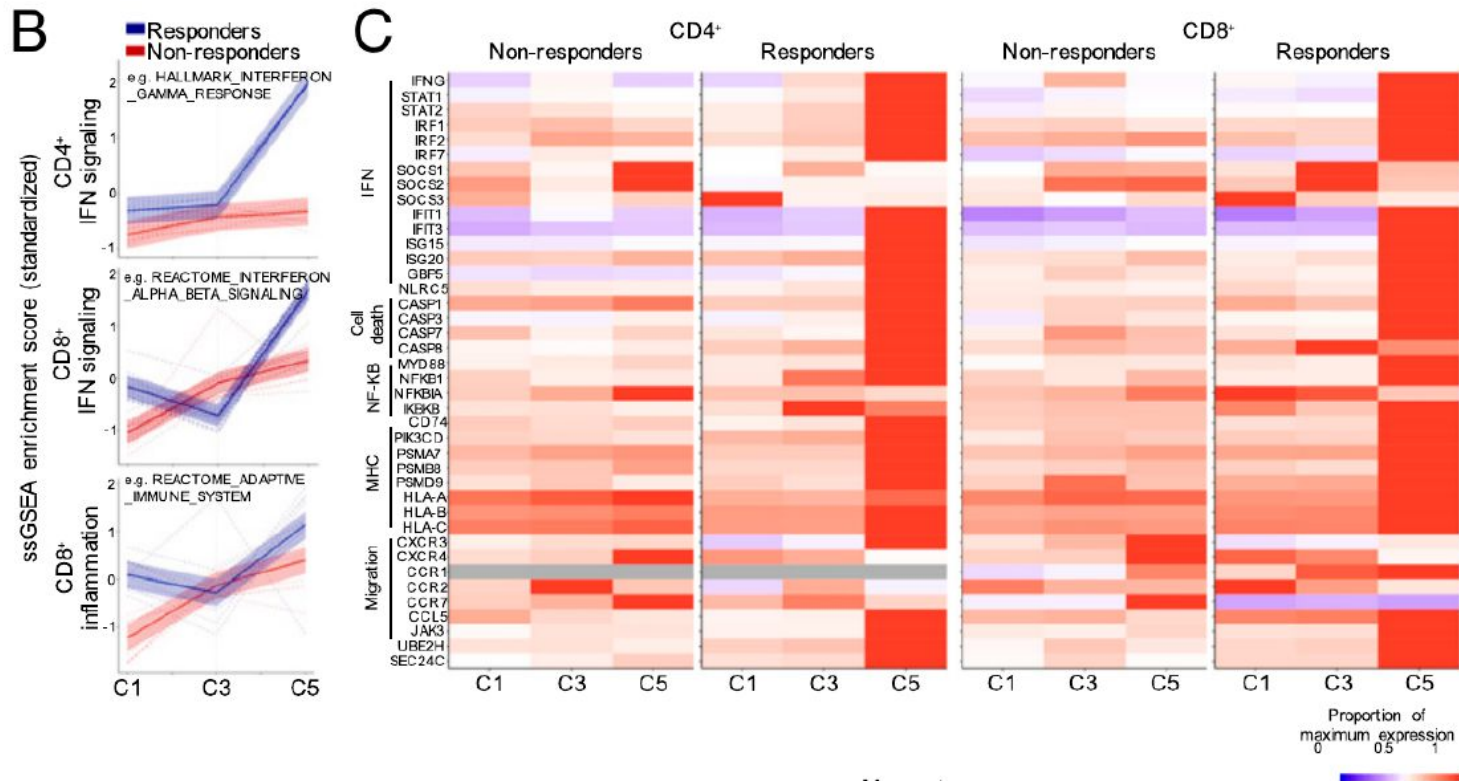
A total number of 70781 cells from 13 patients (7 responders and 6 non-responders) were profiled.

Pathway analysis

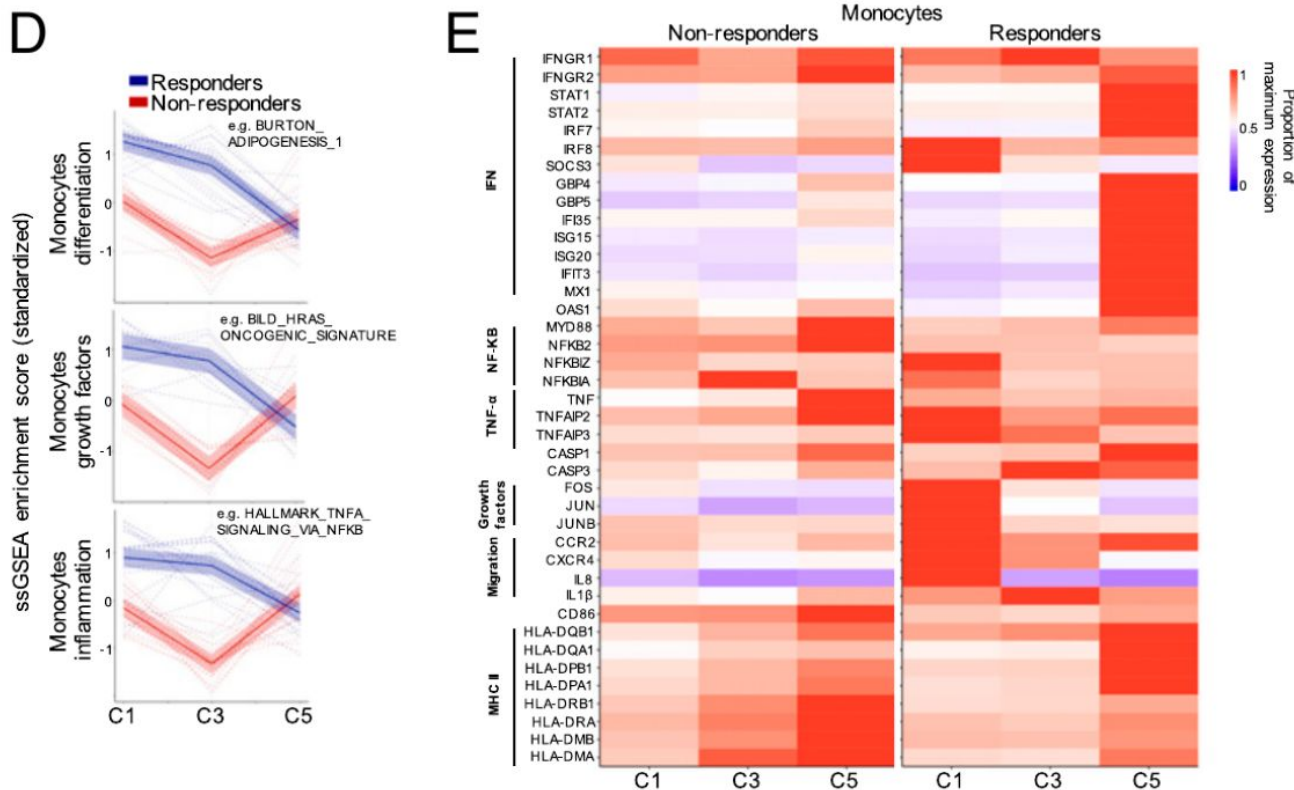
Single-sample gene-set enrichment analysis was performed to identify pathway differences before therapy, during chemotherapy, and during the early combo of chemotherapy and immunotherapy using a *random effects linear model*.



Responders show changes in T-cell signalling during treatment

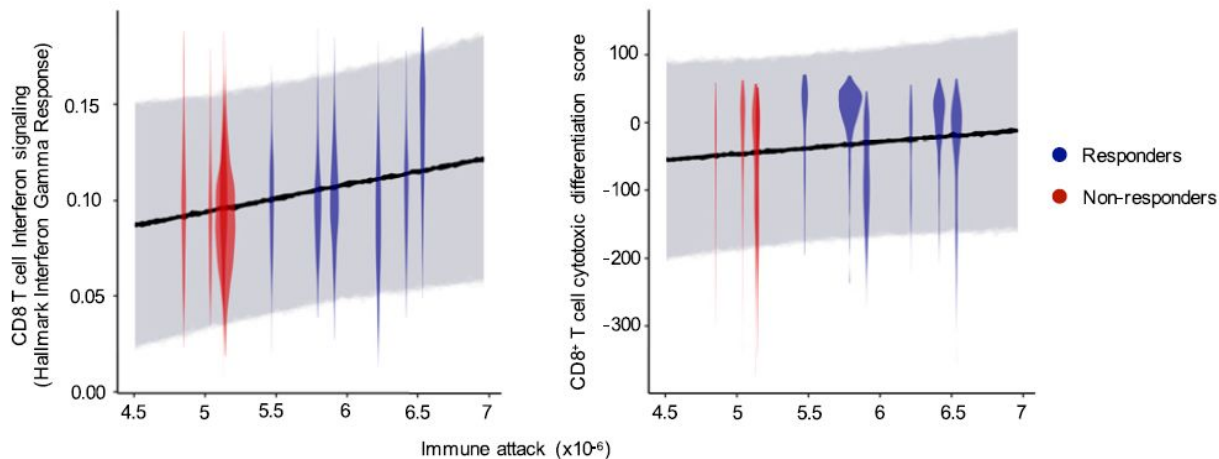


Responders show changes in monocyte signalling during treatment

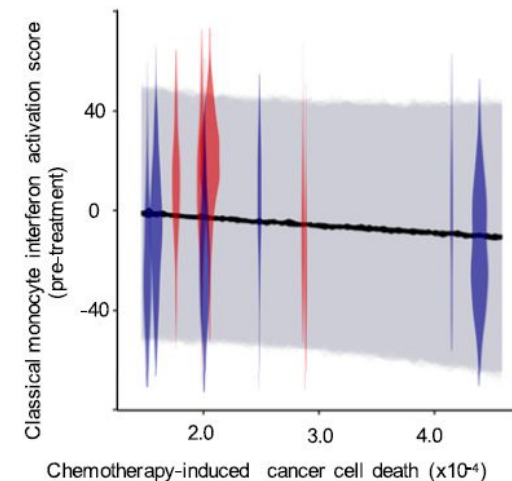


Associations between omics data and inferred model parameter

D



E



Impact and limitation of the study

- *On the biological side*, the results suggest that peripheral blood phenotypes can be used as biomarker of patient responsiveness to therapy. The idea seems to be confirmed the findings by Wu *et al.*, *Peripheral T cell expansion predicts tumour infiltration and clinical response*, Nature 2020.
- *On the modelling side*, the study integrates machine learning, omics data analysis, mathematical modelling techniques to link macroscopic findings, for instance antigen and RECIST scores, with cellular findings, including scRNAseq and flow cytometry. This study exemplifies what we call *multiscale modelling of drug mechanism and safety*.
- We do not know why some patients respond to anti-PD-1 or anti-PDL1 therapies better than other patients based on findings reported in both papers. Nevertheless, both studies suggest that immune cells in peripheral blood may be used as biomarkers in certain settings.

Conclusions

- Understanding how drugs work and how to develop better drugs requires *causal reasoning*, for which there are no scientific consensus yet.
- Integrated mechanistic, computational, and statistical modelling across scales is a viable approach towards causal reasoning.
- Mathematical and computational biology is indispensable to address this grand challenge.

Ways to learn more about mathematical & computational biology in drug discovery

- **People** around you, both with the same and different backgrounds;
- **Reading**, including Journal [Nature Reviews Drug Discovery](#), blogs [In the Pipeline](#), [CureFFI](#), and newsletter [This Week in Mathematical Oncology](#);
- **Online courses**: *Statistical Rethinking* by Richard McElreath, [with freely available lecture videos on YouTube](#), and *Information Theory, Inference, and Learning Algorithms* by David MacKay, [with freely available lecture videos](#).

FDA's opinions on Artificial Intelligence and Machine Learning in Drug Development

1. Human-led governance, accountability, and transparency
2. Quality, reliability, and representativeness of data
3. Model development, performance, monitoring, and validation

That was it, MCBDD 2024

THANK YOU!

References



1. Cristescu, Razvan, Robin Mogg, Mark Ayers, Andrew Albright, Erin Murphy, Jennifer Yearley, Xinwei Sher, et al. 2018. "Pan-Tumor Genomic Biomarkers for PD-1 Checkpoint Blockade–Based Immunotherapy." *Science* 362 (6411): eaar3593. <https://doi.org/10.1126/science.aar3593>.
2. Litchfield, Kevin, James L. Reading, Clare Puttick, Krupa Thakkar, Chris Abbosh, Robert Bentham, Thomas B. K. Watkins, et al. 2021. "Meta-Analysis of Tumor- and T Cell-Intrinsic Mechanisms of Sensitization to Checkpoint Inhibition." *Cell* 184 (3): 596-614.e14. <https://doi.org/10.1016/j.cell.2021.01.002>.
3. Krishnamoorthy, Vijay, Danny J. N. Wong, Matt Wilson, Karthik Raghunathan, Tetsu Ohnuma, Duncan McLean, S. Ramani Moonesinghe, and Steve K. Harris. 2020. "Causal Inference in Perioperative Medicine Observational Research: Part 1, a Graphical Introduction." *British Journal of Anaesthesia* 125 (3): 393–97. <https://doi.org/10.1016/j.bja.2020.03.031>.
4. Krishnamoorthy, Vijay, Duncan McLean, Tetsu Ohnuma, Steve K. Harris, Danny J. N. Wong, Matt Wilson, Ramani Moonesinghe, and Karthik Raghunathan. 2020. "Causal Inference in Perioperative Medicine Observational Research: Part 2, Advanced Methods." *British Journal of Anaesthesia* 125 (3): 398–405. <https://doi.org/10.1016/j.bja.2020.03.032>.
5. Wright, Sewall. 1934. "The Method of Path Coefficients." *The Annals of Mathematical Statistics* 5 (3): 161–215. <https://doi.org/10.1214/aoms/1177732676>.
6. Emdin, Connor A., Amit V. Khera, and Sekar Kathiresan. 2017. "Mendelian Randomization." *JAMA* 318 (19): 1925–26. <https://doi.org/10.1001/jama.2017.17219>.
7. Lawlor, Debbie A., Roger M. Harbord, Jonathan A. C. Sterne, Nic Timpson, and George Davey Smith. 2008. "Mendelian Randomization: Using Genes as Instruments for Making Causal Inferences in Epidemiology." *Statistics in Medicine* 27 (8): 1133–63. <https://doi.org/10.1002/sim.3034>.
8. Lv, Bo-Min, Yuan Quan, and Hong-Yu Zhang. 2021. "Causal Inference in Microbiome Medicine: Principles and Applications." *Trends in Microbiology*, April. <https://doi.org/10.1016/j.tim.2021.03.015>.
9. Smith, George Davey, and Shah Ebrahim. 2008. *Mendelian Randomization: Genetic Variants as Instruments for Strengthening Causal Inference in Observational Studies*. Biosocial Surveys. National Academies Press (US). <https://www.ncbi.nlm.nih.gov/books/NBK62433/>.
10. Williamson, Jon. 2019. "Establishing Causal Claims in Medicine." *International Studies in the Philosophy of Science* 32 (1): 33–61. <https://doi.org/10.1080/02698595.2019.1630927>.

References (continued)

11. Vasudevan, Rama K., Maxim Ziatdinov, Lukas Vlcek, and Sergei V. Kalinin. 2021. "Off-the-Shelf Deep Learning Is Not Enough, and Requires Parsimony, Bayesianity, and Causality." *Npj Computational Materials* 7 (1): 1–6. <https://doi.org/10.1038/s41524-020-00487-0>.
12. Chetty, R. K., J. S. Ozer, A. Lanevski, I. Schuppe-Koistinen, D. McHale, J. S. Pears, J. Vondersch, F. D. Sistare, and F. Dieterle. 2010. "A Systematic Approach to Preclinical and Clinical Safety Biomarker Qualification Incorporating Bradford Hill's Principles of Causality Association." *Clinical Pharmacology & Therapeutics* 88 (2): 260–62. <https://doi.org/10.1038/clpt.2010.77>.
13. Havel, Jonathan J., Diego Chowell, and Timothy A. Chan. 2019. "The Evolving Landscape of Biomarkers for Checkpoint Inhibitor Immunotherapy." *Nature Reviews Cancer* 19 (3): 133–50. <https://doi.org/10.1038/s41568-019-0116-x>.
14. Litchfield, Kevin, James L. Reading, Clare Puttick, Krupa Thakkar, Chris Abbosh, Robert Bentham, Thomas B. K. Watkins, et al. 2021. "Meta-Analysis of Tumor- and T Cell-Intrinsic Mechanisms of Sensitization to Checkpoint Inhibition." *Cell* 184 (3): 596-614.e14. <https://doi.org/10.1016/j.cell.2021.01.002>.
15. Trusheim, Mark R., Ernst R. Berndt, and Frank L. Douglas. 2007. "Stratified Medicine: Strategic and Economic Implications of Combining Drugs and Clinical Biomarkers." *Nature Reviews Drug Discovery* 6 (4): 287–93. <https://doi.org/10.1038/nrd2251>.
16. Trusheim, Mark R., Ernst R. Berndt, and Frank L. Douglas. 2007. "Stratified Medicine: Strategic and Economic Implications of Combining Drugs and Clinical Biomarkers." *Nature Reviews Drug Discovery* 6 (4): 287–93. <https://doi.org/10.1038/nrd2251>.
17. Bender, Andreas, and Isidro Cortés-Ciriano. 2020. "Artificial Intelligence in Drug Discovery: What Is Realistic, What Are Illusions? Part 1: Ways to Make an Impact, and Why We Are Not There Yet." *Drug Discovery Today*, December. <https://doi.org/10.1016/j.drudis.2020.12.009>.
18. Bender, Andreas, and Isidro Cortes-Ciriano. 2021. "Artificial Intelligence in Drug Discovery: What Is Realistic, What Are Illusions? Part 2: A Discussion of Chemical and Biological Data." *Drug Discovery Today* 26 (4): 1040–52. <https://doi.org/10.1016/j.drudis.2020.11.037>.
19. Robers, M.B., R. Friedman-Ohana, K.V.M. Huber, L. Kilpatrick, J.D. Vasta, B.-T. Berger, C. Chaudhry, et al. 2020. "Quantifying Target Occupancy of Small Molecules Within Living Cells." *Annual Review of Biochemistry* 89 (1): 557–81. <https://doi.org/10.1146/annurev-biochem-011420-092302>.
20. Simon, Gabriel M., Micah J. Niphakis, and Benjamin F. Cravatt. 2013. "Determining Target Engagement in Living Systems." *Nature Chemical Biology* 9 (4): 200–205. <https://doi.org/10.1038/nchembio.1211>.
21. Grimwood, Sarah, and Paul R. Hartig. 2009. "Target Site Occupancy: Emerging Generalizations from Clinical and Preclinical Studies." *Pharmacology & Therapeutics* 122 (3): 281–301. <https://doi.org/10.1016/j.pharmthera.2009.03.002>

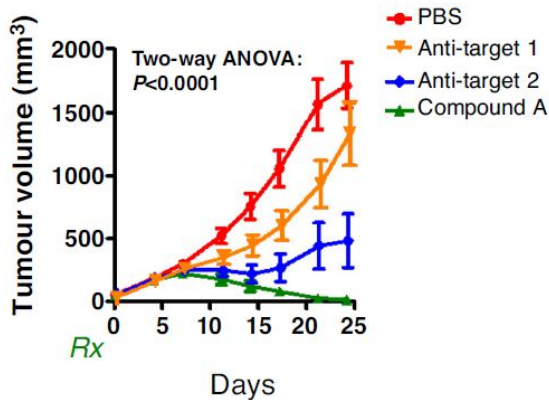
References (continued)

22. Woude, Lieke L. van der, Mark A. J. Gorris, Altuna Halilovic, Carl G. Figdor, and I. Jolanda M. de Vries. 2017. "Migrating into the Tumor: A Roadmap for T Cells." *Trends in Cancer* 3 (11): 797–808. <https://doi.org/10.1016/j.trecan.2017.09.006>.
23. Griffiths, Jason I., Pierre Wallet, Lance T. Pflieger, David Stenehjem, Xuan Liu, Patrick A. Cosgrove, Neena A. Leggett, et al. 2020. "Circulating Immune Cell Phenotype Dynamics Reflect the Strength of Tumor–Immune Cell Interactions in Patients during Immunotherapy." *Proceedings of the National Academy of Sciences*, June. <https://doi.org/10.1073/pnas.1918937117>.
24. Lotka-Volterra equation visualization: https://upload.wikimedia.org/wikipedia/commons/d/d7/Lotka_Volterra_equation_Maple_plot.png, adapted due to possible mislabelling
25. MacKay, David J. C. 2003. *Information Theory, Inference, and Learning Algorithms*. Cambridge, UK ; New York: Cambridge University Press. <http://www.inference.org.uk/mackay/itila/book.html>.
26. McElreath, Richard. 2020. *Statistical Rethinking: A Bayesian Course with Examples in R and Stan*. 2nd ed. CRC Texts in Statistical Science. Boca Raton: Taylor and Francis, CRC Press.
27. Macnamara, Cicely K. 2021. "Biomechanical Modelling of Cancer: Agent-Based Force-Based Models of Solid Tumours within the Context of the Tumour Microenvironment." *Computational and Systems Oncology* 1 (2): e1018. <https://doi.org/10.1002/cso2.1018>.
28. Gündner, Anna Lisa, Gonzalo Duran-Pacheco, Silke Zimmermann, Iris Ruf, Tim Moors, Karlheinz Baumann, Ravi Jagasia, Wilma D. J. van de Berg, and Thomas Kremer. 2019. "Path Mediation Analysis Reveals GBA Impacts Lewy Body Disease Status by Increasing α -Synuclein Levels." *Neurobiology of Disease* 121 (January): 205–13. <https://doi.org/10.1016/j.nbd.2018.09.015>.
29. Wright, Sewall. 1920. "The Relative Importance of Heredity and Environment in Determining the Piebald Pattern of Guinea-Pigs." *Proceedings of the National Academy of Sciences* 6 (6): 320–32. <https://doi.org/10.1073/pnas.6.6.320>.
30. Pearl, Judea. 2009. "Causal Inference in Statistics: An Overview." *Statistics Surveys* 3: 96–146. <https://doi.org/10.1214/09-SS057>.

References (continued)

31. Krishnamoorthy, Vijay, Danny J. N. Wong, Matt Wilson, Karthik Raghunathan, Tetsu Ohnuma, Duncan McLean, S. Ramani Moonesinghe, and Steve K. Harris. 2020. "Causal Inference in Perioperative Medicine Observational Research: Part 1. a Graphical Introduction." *British Journal of Anaesthesia* 125 (3): 393–97. <https://doi.org/10.1016/j.bja.2020.03.031>.
32. Krishnamoorthy, Vijay, Duncan McLean, Tetsu Ohnuma, Steve K. Harris, Danny J. N. Wong, Matt Wilson, Ramani Moonesinghe, and Karthik Raghunathan. 2020. "Causal Inference in Perioperative Medicine Observational Research: Part 2, Advanced Methods." *British Journal of Anaesthesia* 125 (3): 398–405. <https://doi.org/10.1016/j.bja.2020.03.032>.
33. Senn, Stephen. 2018. "Statistical Pitfalls of Personalized Medicine." *Nature* 563 (7733): 619. <https://doi.org/10.1038/d41586-018-07535-2>.
34. Xu, Chao, Patanjali Ravva, Jun Steve Dang, Johann Laurent, Céline Adessi, Christine McIntyre, Georgina Meneses-Lorente, and François Mercier. 2018. "A Continuous-Time Multistate Markov Model to Describe the Occurrence and Severity of Diarrhea Events in Metastatic Breast Cancer Patients Treated with Lumretuzumab in Combination with Pertuzumab and Paclitaxel." *Cancer Chemotherapy and Pharmacology* 82 (3): 395–406. <https://doi.org/10.1007/s00280-018-3621-9>.
35. Kather, Jakob Nikolas, Jan Poleszczuk, Meggy Suarez-Carmona, Johannes Krisam, Pornpimol Charoentong, Nektarios A. Valous, Cleo-Aron Weis, et al. 2017. "In Silico Modeling of Immunotherapy and Stroma-Targeting Therapies in Human Colorectal Cancer." *Cancer Research* 77 (22): 6442–52. <https://doi.org/10.1158/0008-5472.CAN-17-2006>.
36. Beckers, Thomas. "An Introduction to Gaussian Process Models." arXiv, February 10, 2021. <https://doi.org/10.48550/arXiv.2102.05497>.
37. <https://www.fda.gov/media/167973/download>
- 38.

A real-word example with a bispecific antibody



Anti-target 1 TO
required for
efficacy in tumour
and PBMCs

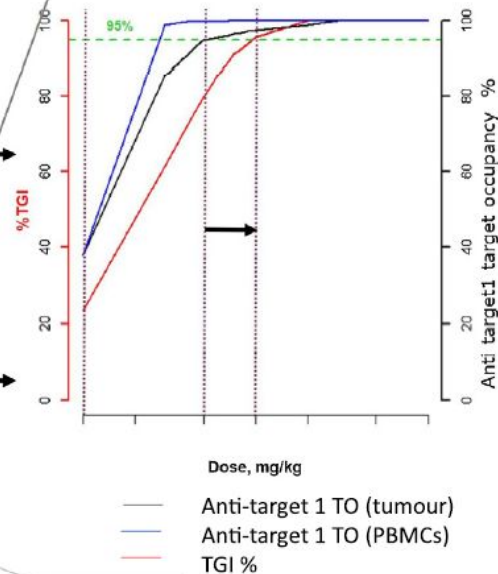
Efficacious
concentration

Pharmacokinetics
Intravenous PK

PK/PD model
Anti-target1 TO in EMT-6
model

PK/efficacy model
Tumour growth model
in EMT-6 model

Predicted human PK



TGI: tumour growth inhibition; TO: target occupancy;
PBMC, peripheral blood mononuclear cells

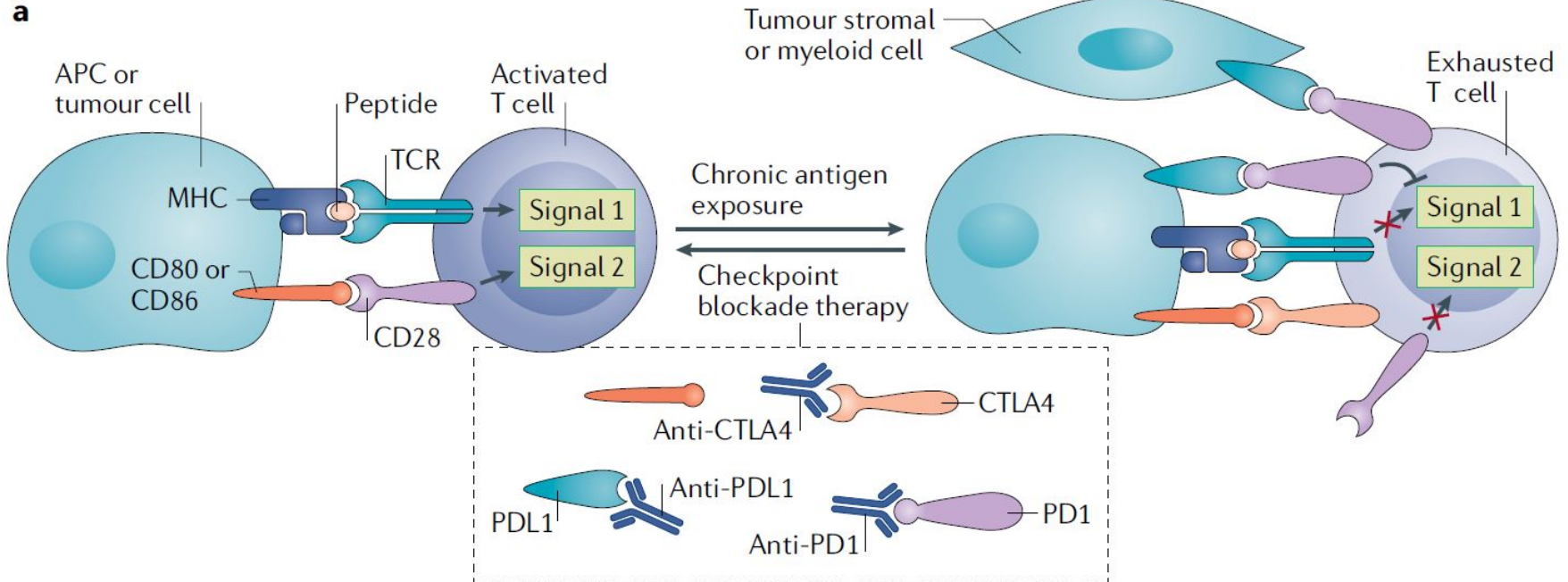
Exposure-response in animal model and translatable biomarkers are essential for dose prediction

TABLE 2

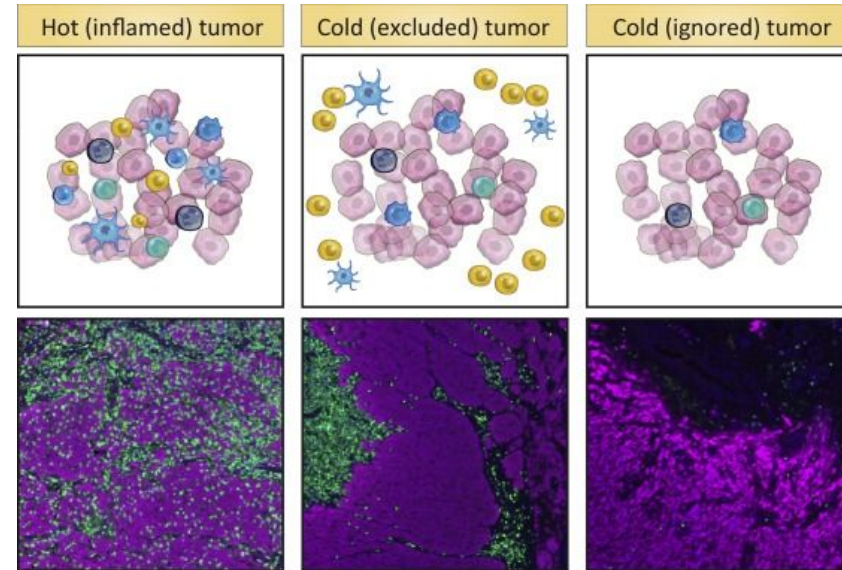
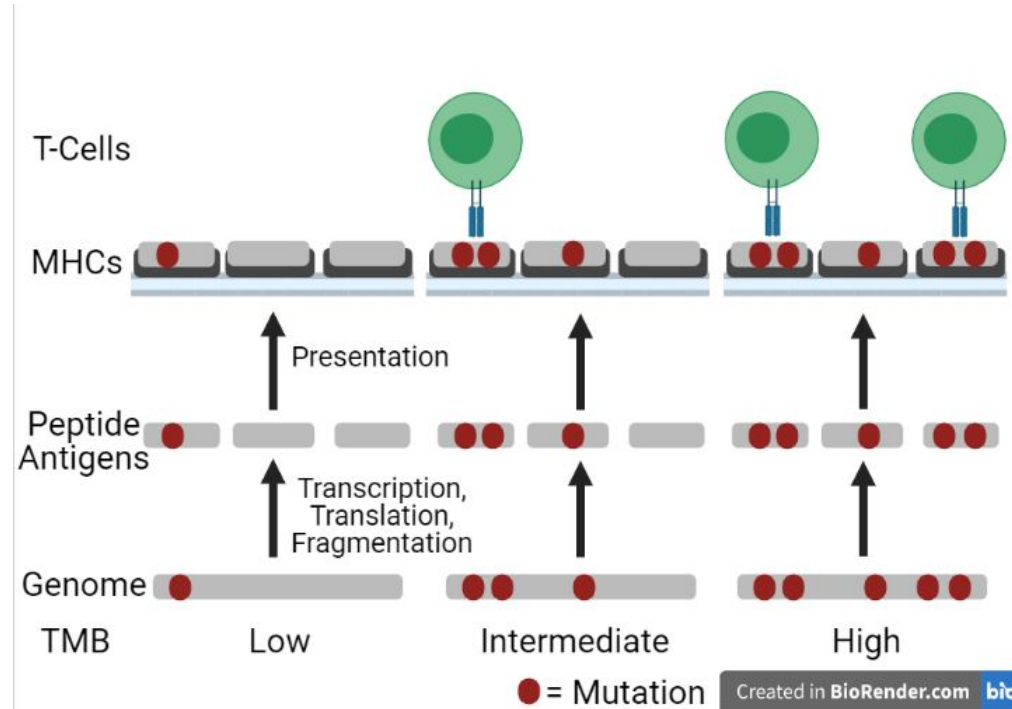
Correlation of responses to dose-related questions (Q) of TmX Guide to dose prediction successes or observation of efficacy in the clinic

Category	Q1: desired exposure–response in appropriate animal model?	Q2: Translatable biomarkers?	Number of drugs for which model-based active dose prediction is within twofold or clinical efficacy is observed within predicted dose range out of total number in category
1	Yes	Yes	5/6
2 ^a	No	No	1/6
3	No	Yes	2/2
4 ^b	Yes	No	0/1

Molecular basis of cancer immunotherapy



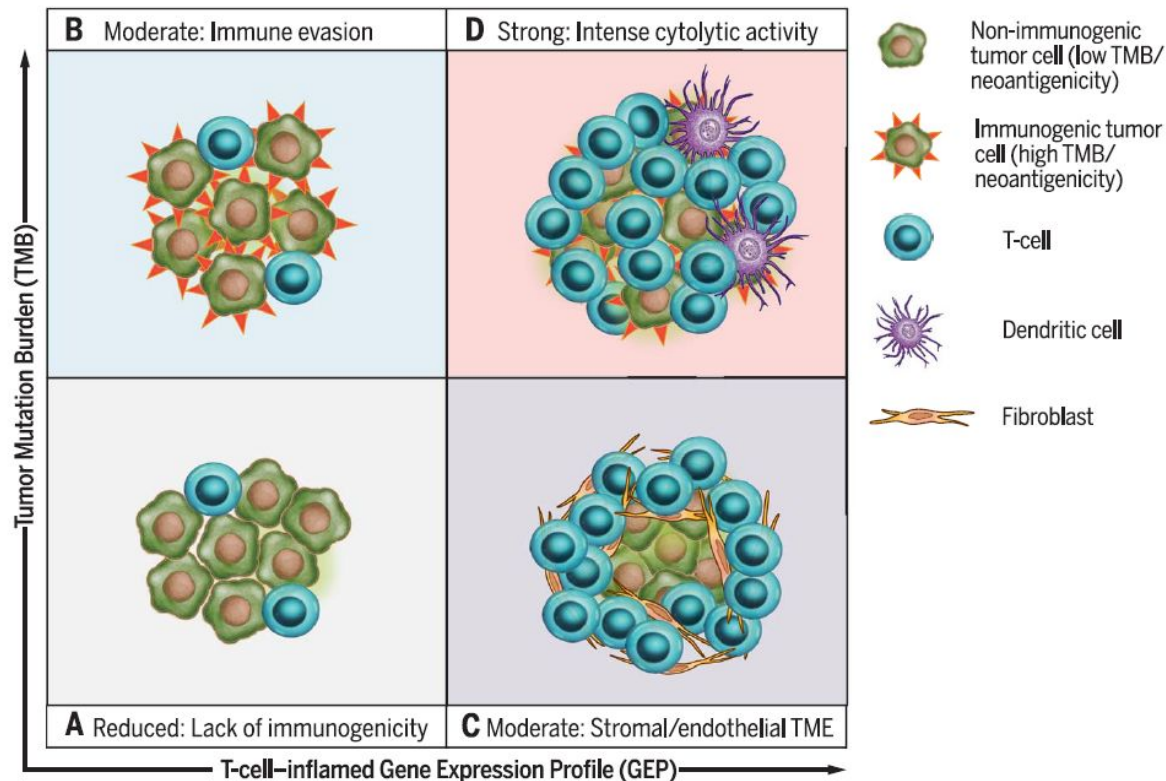
Tumour mutation burden and immune phenotype may affect the effect of immunotherapy



Trends in Cancer

Cristescu *et al.* established TMB and T-cell-inflamed Gene Expression Profile (GEP) as biomarkers

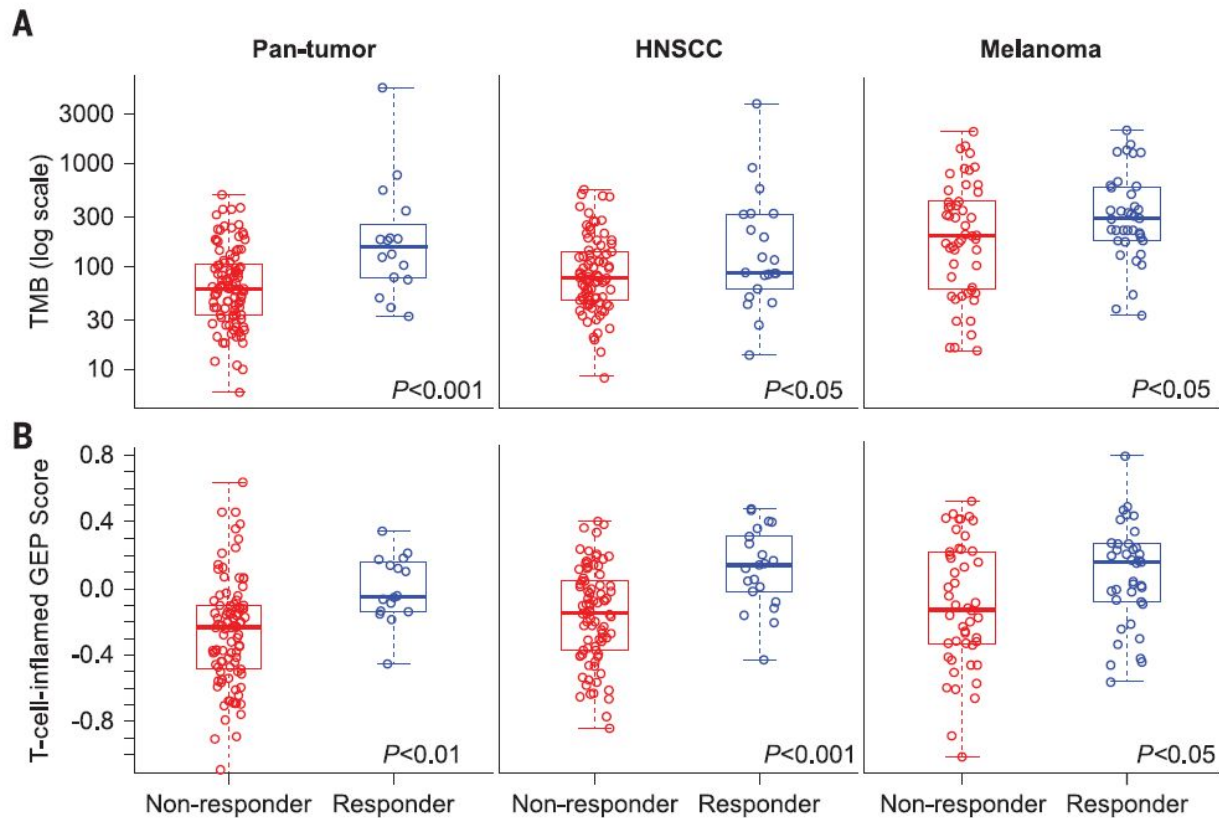
Patients with high tumor mutation burden **AND** a T-cell-inflamed gene expression profile (TME) are more likely to respond to cancer immunotherapy.



Univariate analysis establishes correlation between TMB/GEP and responsiveness

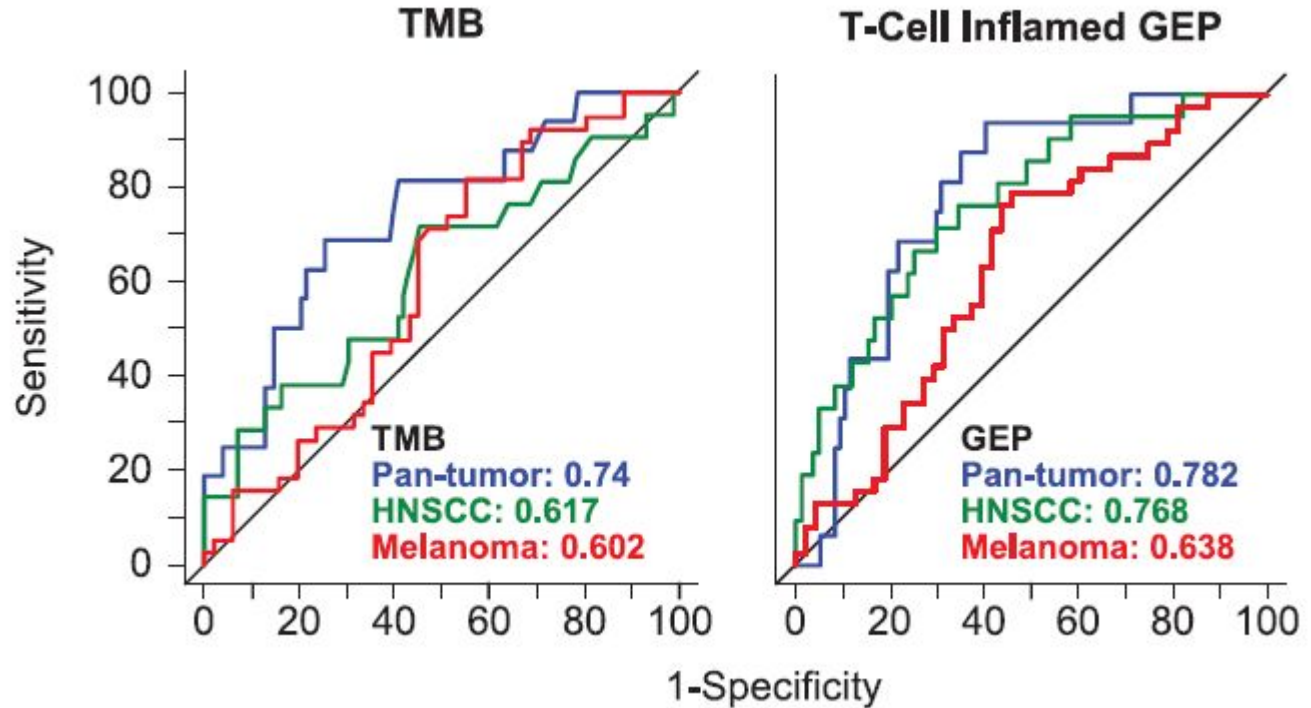
GEP: weighted sum of normalized expression of 18 genes related with immune response (CCL5, CD27, CD274 (PD-L1), CD276 (B7-H3), CD8A, CMKLR1, CXCL9, CXCR6, HLA-DQA1, HLA-DRB1, HLA-E, IDO1, LAG3, NKG7, PDCD1LG2 (PDL2), PSMB10, STAT1, and TIGIT).

HNSCC: head and neck cancer



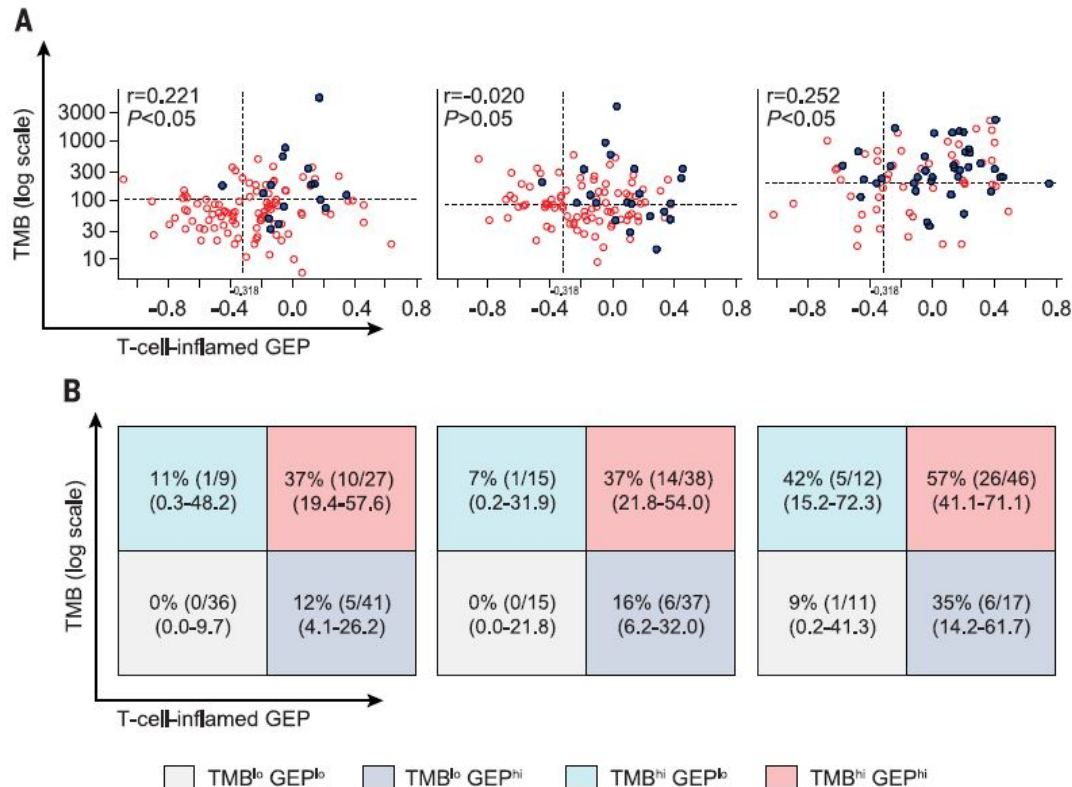
Both TMB and GEP can partially predict responsiveness

Receiver
Operating
Characteristic
(ROC) curves of
using either TMB
or GEP for binary
classification.
Metrics: Area
Under ROC
(AUROC)

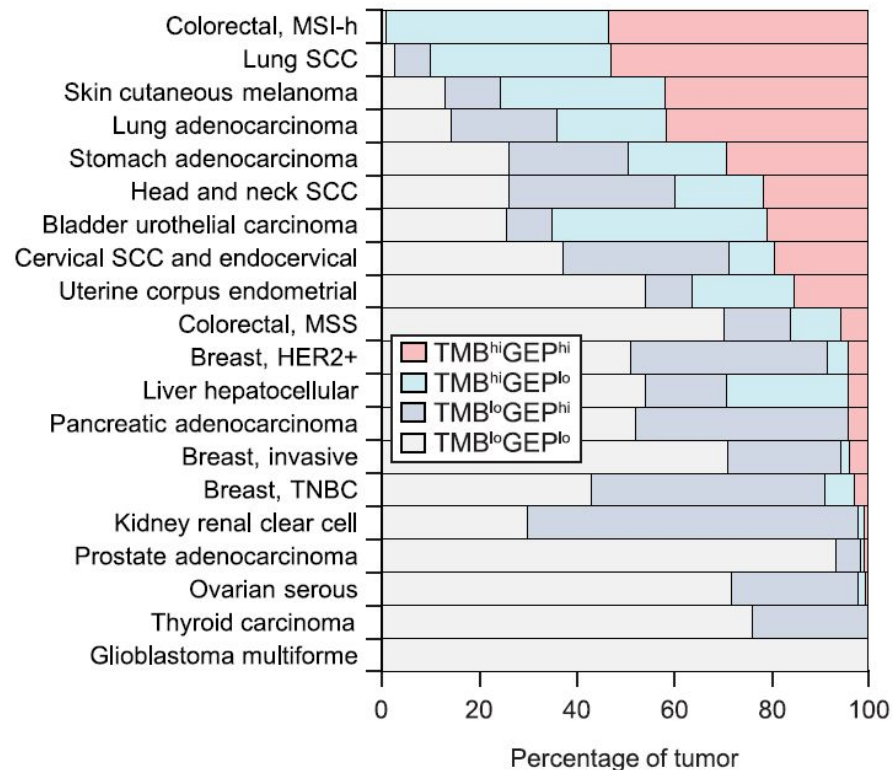
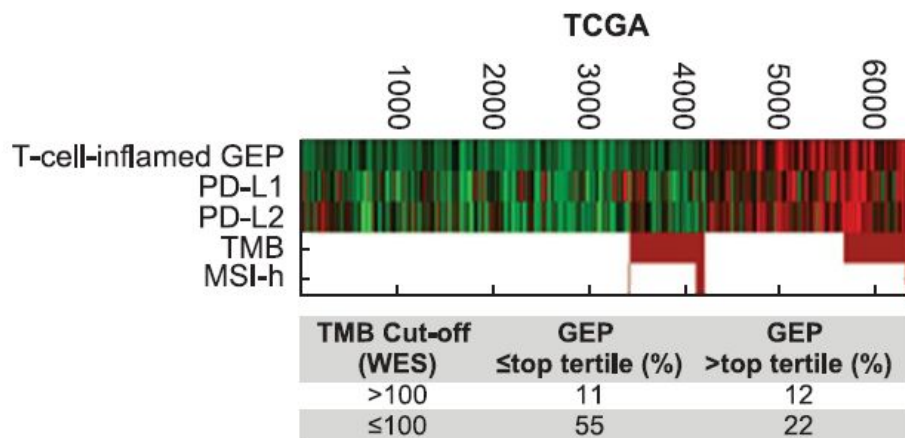


High TMB and high GEP are associated with higher responsiveness to anti-PD1 antibody treatment

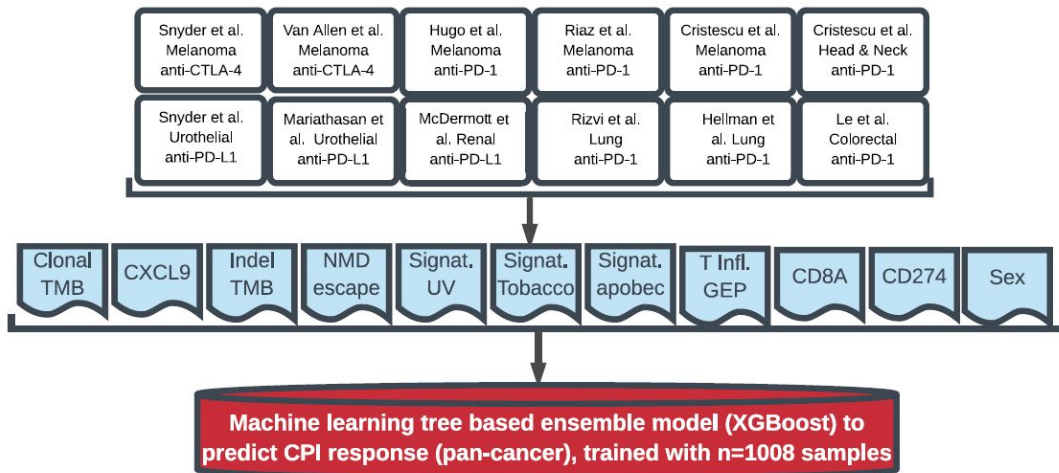
- From left to right: three patient cohorts (pan-cancer; head-and-neck cancer; melanoma)
- Open red circles: non responders; Black dots: responders.



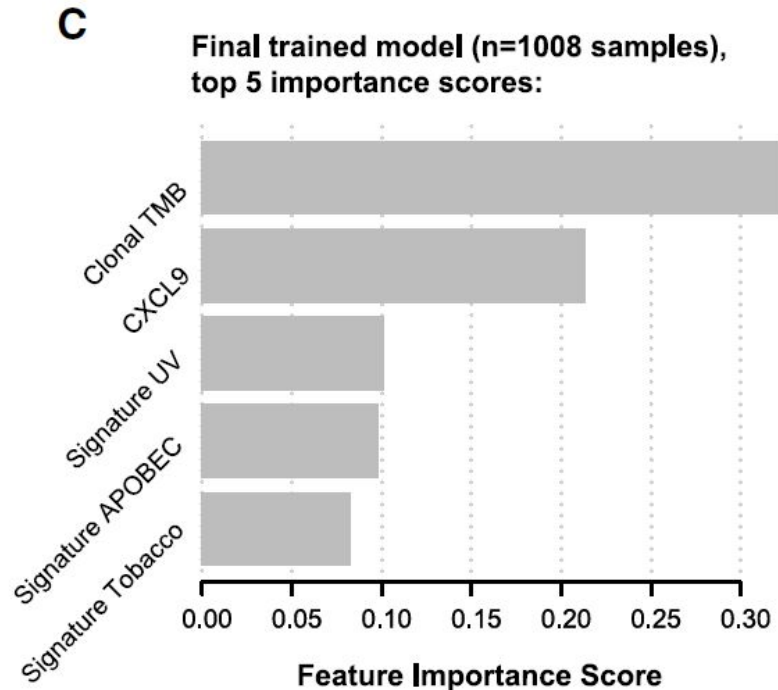
Data mining in public cancer database TCGA suggests potential indications



Meta-analysis (Litchfield *et al.* 2021) confirms TMB and T-cell infiltration as predictors of responsiveness

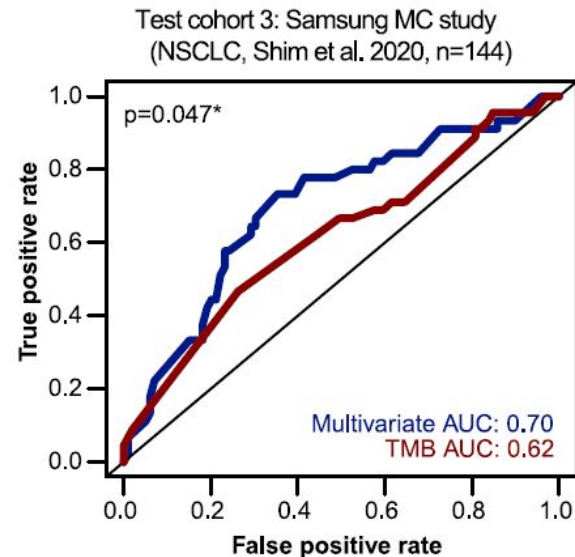
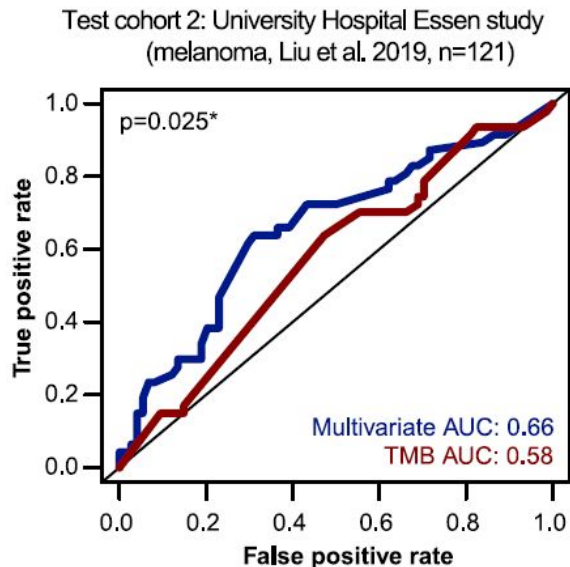
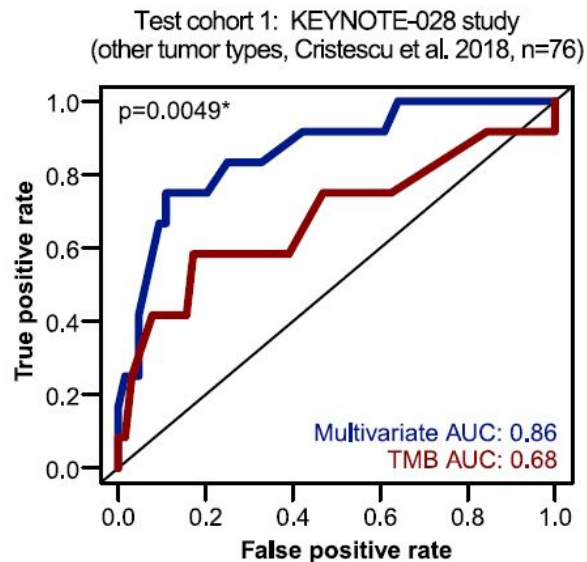


CXCL9 is a chemokine that enhances recruitment of cytotoxic CD8⁺ T cells into the tumor.



The multivariate classifier improves performance, but to predict responsiveness is an open question

Testing of TMB versus multivariable CPI stratifier performance in three independent test cohorts (total n=341):



-
- a** Renal2 $n_D = 139$ $r_w = 0.93$ Renal1 $n_D = 108$ $r_w = 0.81$ Renal3 $n_D = 119$ $r_w = 0.58$ Lung6 $n_D = 213$ $r_w = 0.31$
- Y-axis: Tumor cell fraction
X-axis: NAT cell fraction
- a** scRNA-seq ($n = 141,623$) **b** scTCR-seq ($n = 89,319$)
- Clone size, blood
 • 0 • 10
 • 1 • 20
 • 2 • 50
 • 5 • 100
- Clone size, tumour
 0 1 ∞
 0 1 ∞
- Clone size, NAT
 0 1 ∞
- Blood singleton
 Blood multiplet

Bonus: Mathematical modelling of epidemiology

The SIR (S =susceptible, I =infectious, R =removed) model modelling epidemiology (without viral dynamics, $N = S + I + R$).

$$\frac{dS}{dt} = -\frac{\beta IS}{N}, \quad (3)$$

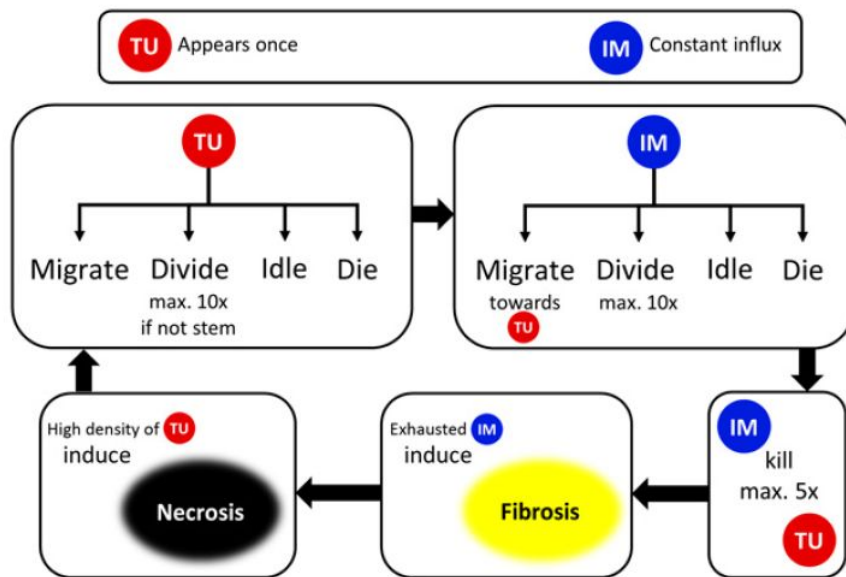
$$\frac{dI}{dt} = \frac{\beta IS}{N} - \gamma I, \quad (4)$$

$$\frac{dR}{dt} = \gamma I \quad (5)$$

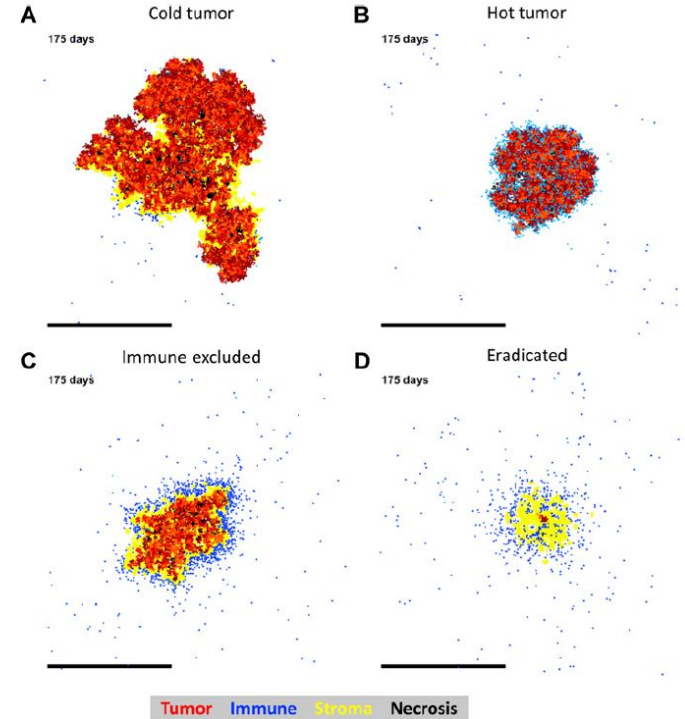
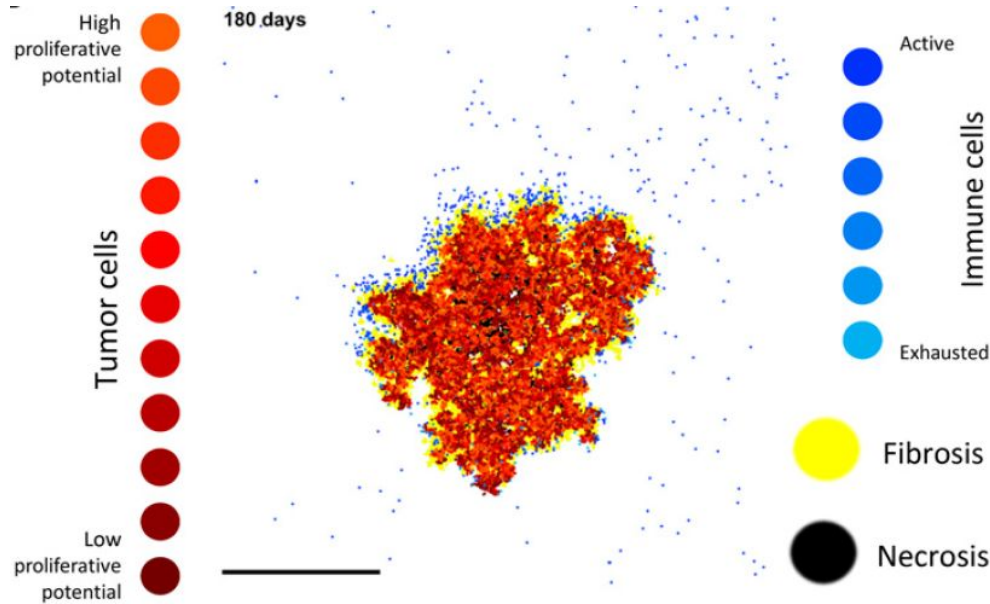
Agent-based modelling of cancer immunotherapy for colorectal cancer without high TMB

Table 1. Assumptions for the model and references for each assumption

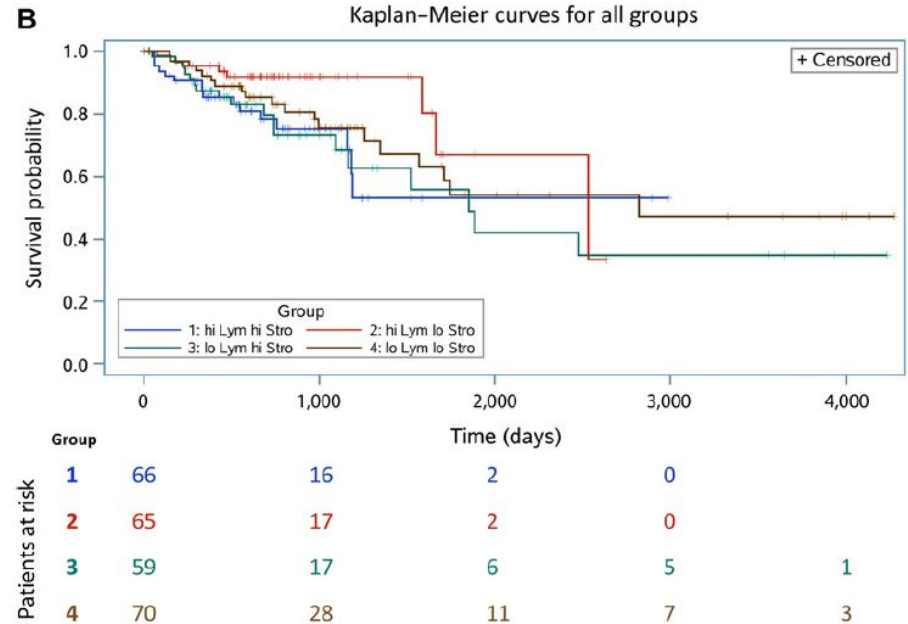
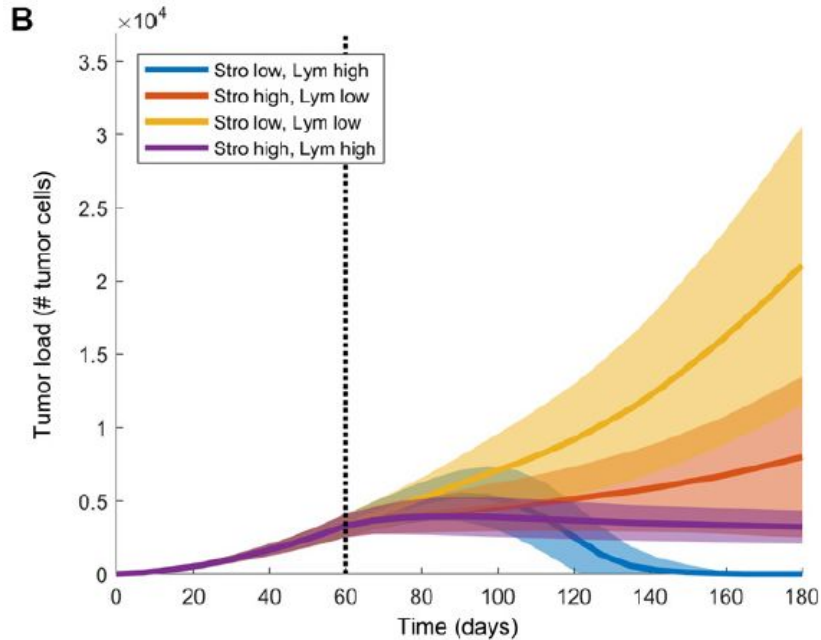
Assumption	Ref.
All cells can migrate, proliferate, and die.	Trivial
Tumor cells are composed of stem cells and non-stem cells. Stem cells can divide symmetrically with a fixed probability.	(14)
Stem cells can proliferate indefinitely, all other cells die after a fixed number of proliferation cycles.	(14)
All cells can spontaneously enter apoptosis.	Own data
Tumor cells can spontaneously enter necrosis.	Own data
Tumor cells that are far from the outer margin have a higher probability of entering necrosis than those cells closer to the margin.	Own data
Immune cells are generated through a steady influx into the domain and proliferation within the domain.	(32), own data
Immune cells move by a "random walk" but have a tendency to migrate toward tumor cells.	(31–33), own data
Immune cells can kill adjacent tumor cells whenever they are close enough. Killing, like other events in the model, occurs stochastically with a fixed probability and is not regulated by other factors.	(23)
Immune cells can kill five times before they become exhausted, which means that they cannot kill anymore but can still proliferate.	(23, 34)
Activated immune cells give rise to stroma through a desmoplastic reaction (stroma reaction). For simplicity, this behavior is restricted to immune cells that have successfully killed five times in the model.	(35, 36)
By default, cells cannot migrate through stroma, but stromal permeability can be increased optionally.	(37)



Agent-based modelling of cancer immunotherapy for colorectal cancer without high TMB



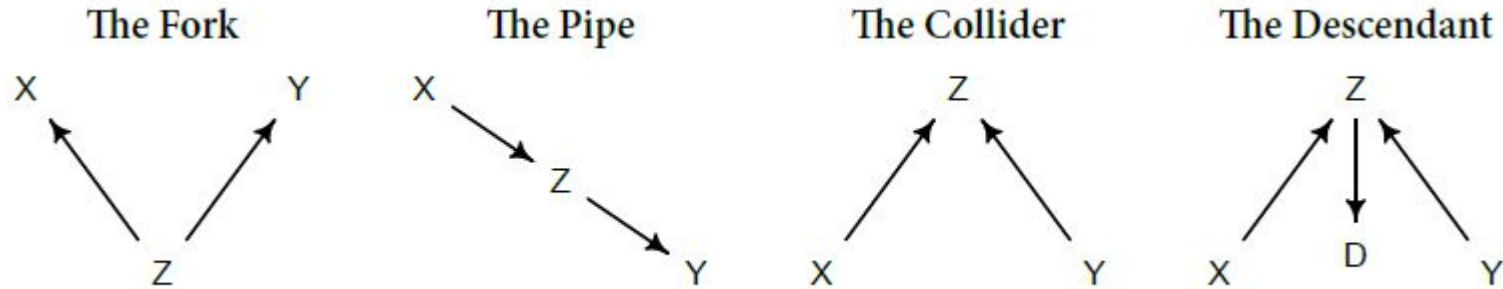
Counterfactual and statistical analysis allow us *learn* from the models *confirm* the learnings



Bradford Hill Criteria for causation

1. **Strength** (effect size)
2. **Consistency** (reproducibility)
3. ***Specificity***
4. **Temporality**
5. **Biological gradient** (dose-response relationship)
6. ***Plausibility***
7. **Coherence**
8. **Experiment**
9. **Analogy** (similarity)
10. **Reversibility** (proposed by others)

Statistical causal inference with Directed Acyclic Graphs (DAGs)



Reading: chapter 1-6 of *Statistical Rethinking* (2nd Edition)